

Introduction to fixed prosthodontics

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Objective

- Introducing fixed prosthodontics (Inlay, Onlay, Veneers, Crown, and bridge)
- To highlight students with basic knowledge on contemporary practice of fixed prosthodontics
- To highlight students with basic skills in contemporary practice of fixed prosthodontics

Q?

- The difference between dentures and over dentures
- The difference between over denture versus implants
- What is Chromo cobalt (cobalt chromo)



Complete Denture



Flexible Denture



Precision Attachment



Flipper Denture



Q?

❖ Describe the following terminologies ?

➤ Inlay

➤ On lay

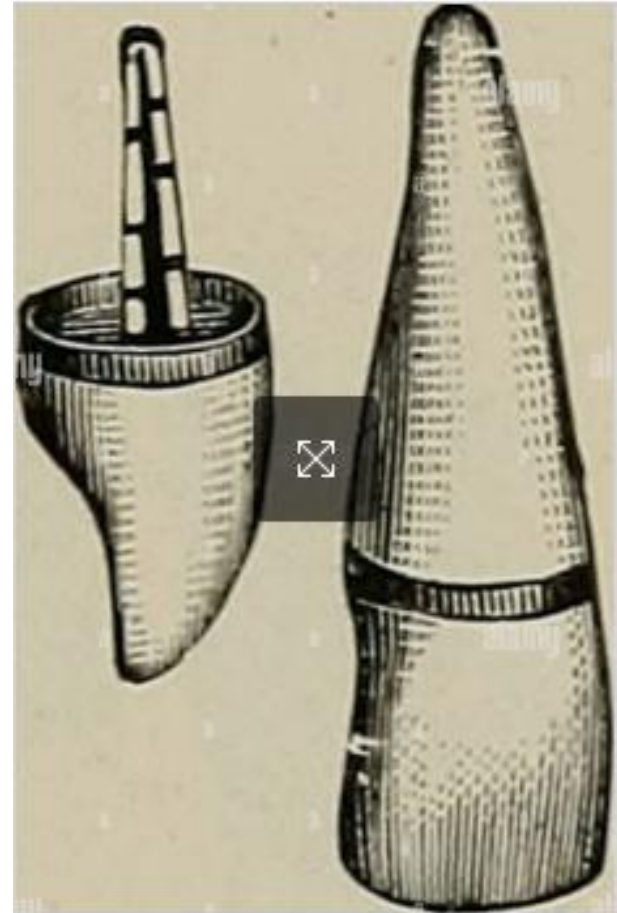
➤ Crown (Richmond, cantilever, pontic)

➤ Veneers

➤ Bridges

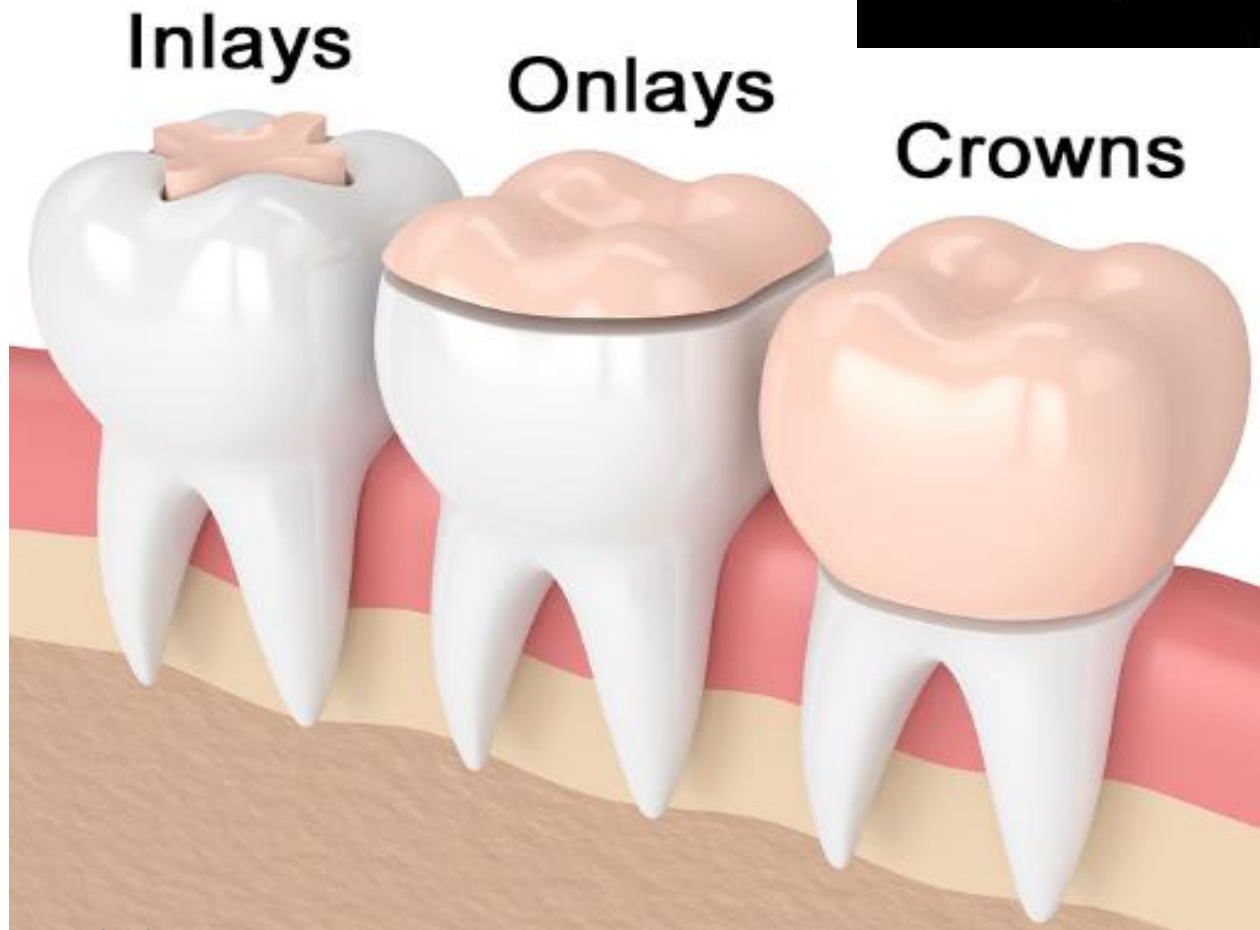


Richmond Crown



Veneers





Materials for Prosthesis construction

- Metals (Cobalt chromium, gold, iron, nickel, silver, titanium, zirconium)
- Ceramic (porcelain, glass ceramic, e-max, silicon carbide)
- Acrylic
- ❖ NB: fixed prosthodontic crowns are constructed either from metal or ceramic materials or the combination of the two.

Q?

❖ Describe the following terminologies ?

➤ Metal and Alloy

➤ Ceramic

➤ Zirconia

➤ E-max

➤ Acrylic

Metals

- *Metal, is any class of substances characterized by high electrical and thermal conductivity as well as malleability, ductility, and high reflectivity of light.*
- *Metal in dentistry is used either in pure form or in form of **alloys**.*
- ***Alloys*** : *a material made by mixing two or more metals or by combining metal with non metals to improve certain properties like strength, durability, corrosion resistance, or appearance.*

Metals



Cont`d

➤ *Metals used for crown construction*

- ✓ Gold and Gold alloy
- ✓ Silver palladium alloy
- ✓ Metal ceramic alloy

➤ *Metals used for RPD construction*

- ✓ Cobalt chromium
- ✓ Nickel chromium
- ✓ Aluminum alloys
- ✓ Type IV gold alloys and titanium.

Metal-ceramic alloy

- They are alloys that are compatible with porcelain and capable of bonding to it, a layer of porcelain is fused to the alloy to give it natural tooth like appearance.
- Porcelain is brittle but these alloys gives it reinforcement.
- These alloys should have coefficient of thermal expansion (C.T.E.) which matches that of porcelain.



Ceramic

- The word Ceramic is derived from the Greek word “keramos”, which literally means ‘**burnt stuff**’.
- A ceramic is an earthly material usually of silicate nature and may be defined as a combination of one or more metals like aluminum, calcium, lithium, magnesium, silicon, zirconium & titanium with a non-metallic element usually oxygen. Example : aluminium & oxygen (alumina - Al_2O_3), calcium & oxygen (calcia - CaO), silicon & nitrogen (nitride- Si_3N_4) ([American ceramic society](#))

Major constituents of ceramics

Koalin , hydrated aluminum silicate



Quartz, silica dioxide



Feldspar, potassium
aluminum silicate

Composition of ceramics

Ingredients	Functions
Feldspar (naturally occurring minerals composed of potash [K ₂ O], soda [Na ₂ O], alumina and silica).	It is the lowest fusing component, which melts first and flows during firing, initiating these components into a solid mass.
Quartz (silica)	• Strengthens the fired porcelain restoration. • Remains unchanged at the temperature normally used in firing porcelain and thus contribute stability to the mass during heating by providing framework for the other ingredients.
Kaolin (Al ₂ O ₃ .2 SiO ₂ . 2H ₂ O - Hydrated aluminosilicates)	Used as a binder; Increases moldability of the unfired porcelain; Imparts opacity to the finished porcelain product.
Glass modifiers, e.g. K, Na, or Ca oxides or basic oxides	They interrupt the integrity of silica network and acts as flux.
Color pigments or frits, e.g. Fe/Ni oxide, Cu oxide, MgO, TiO ₂ , and Co oxide.	To provide appropriate shade to the restoration.

Types of ceramics for dental use

➤ Based on the process used for strengthening and the major **constitute** used for strengthening they can be named:

✓ Porcelain

✓ Zirconium

✓ E-max or others.....

Ceramics



Porcelain : A glass based ceramic usually made up of white clay



Bentonite

French Green

White Cosmetic

Rhassoul

Fuller's Earth

Zirconium

- Zirconia (ZrO_2) is a crystalline oxide of **zirconium** which holds good mechanical, optical, and biological properties. This biomaterial has three basic chemical forms; monoclinic, tetragonal, and cubic.



E-max

- E –max crown is a type of all ceramic crown which is preferred for its long lasting and aesthetic qualities
- Crown is made from a single block of **Lithium disilicate** ceramic. This is top grade material which has been harvested for its toughness, durability and opaque qualities which makes it a highly prized crown.
- The crown is considered to be the best match with your own natural teeth.

TYPES OF CROWNS



Full metal



Porcelain fused to metal



Zirconia



All-ceramic / All-porcelain



Zirconium Porcelain Crowns



E Max Crowns



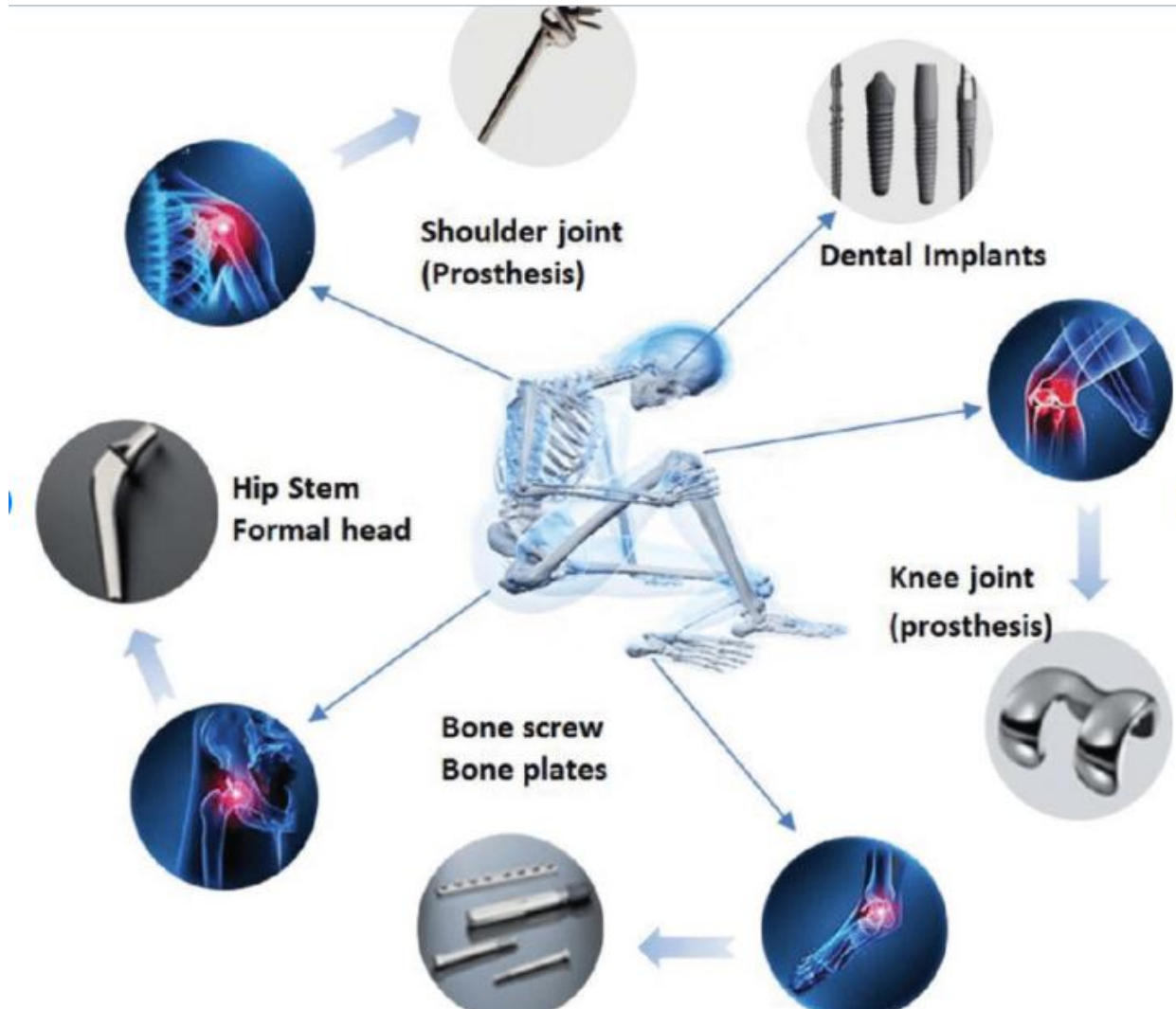
Porcelain Fused Metal Crowns



Q?

➤ Implant

What is that?



Dental implant

- A **dental implant** (also known as an **endosseous implant** or **fixture**) is a prosthesis that interfaces with the bone of the jaw or skull to support a dental prosthesis such as a crown, bridge, denture, or facial prosthesis or to act as an orthodontic anchor.



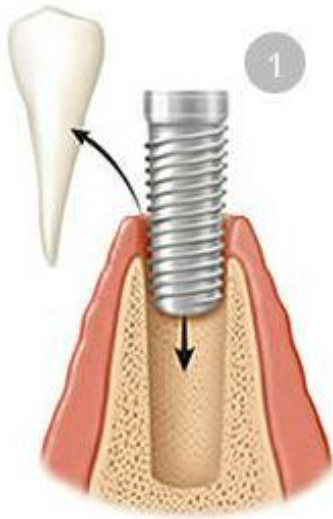
Dental Implant Screws



Dental Implant Custom Abutment Types



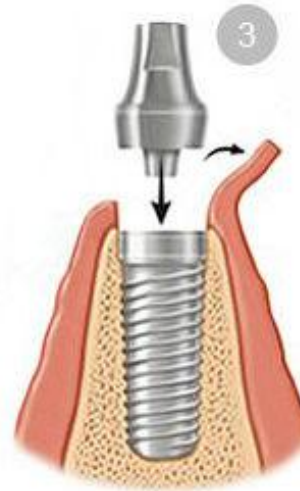
Dental Implantation Steps



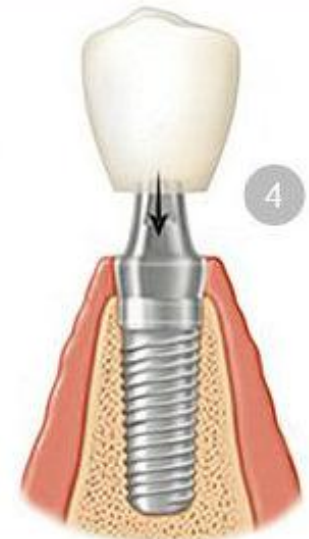
Implant is inserted into the bone



Healing process of the bone (osseointegration)

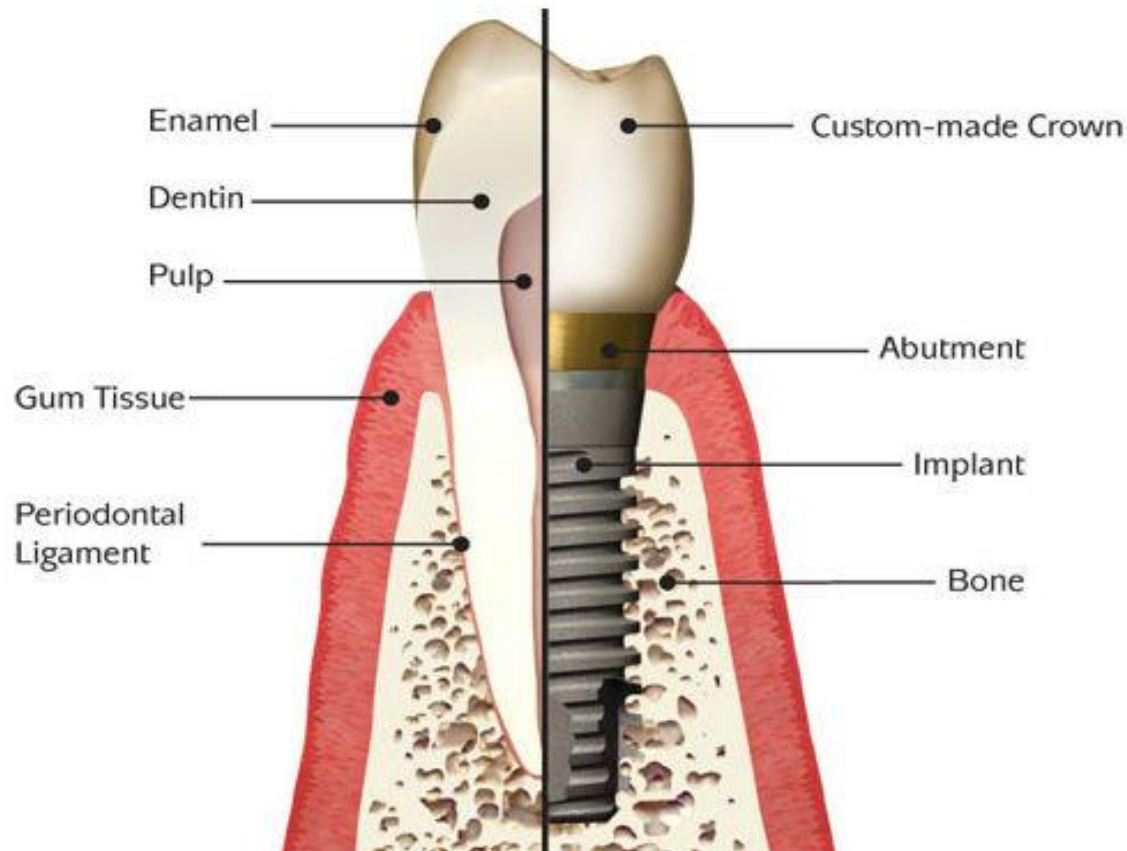


Dental abutment is placed on the dental implant



Ceramic crown is placed which replaces the real tooth

Healthy Tooth vs. Dental Implant



Clinical steps of fixed prosthesis construction

- History, P/E and selection of prospective abutment
- Treatment planning for fixed prosthodontics
- Tooth preparation
- Impression taking
- Shade selection
- Prosthodontic request prescription
- Cementation
- Occlusion verification
- Complications following fixed prosthodontics

History and physical examination in fixed prosthodontics

- **Diagnosis** is simply defined as the procedure of collecting data and information through different channels as that a proper line of treatment could be proposed.
- Components of proper Dx are:
 - 1) History (dental and medical).
 - 2) TMJ/occlusal evaluation
 - 3) Intra oral examination.
 - 4) Diagnostic cast analysis.
 - 5) Full mouth radiographic evaluation.

A. History

- 1) Personal details: The patient's name, address, phone number, sex, occupation, work schedule, and marital and financial status are noted.
- 2) Dental history: includes *Patient's chief complaint* ; Previous treatments and patient's attitude and Patient's expectation's from treatment.
- 3) Medical history : chronic diseases with oral manifestations; allergic rxn; hx of medications; hx of highly contagious infectious diseases and history of pervious radiation therapy

B. Examination

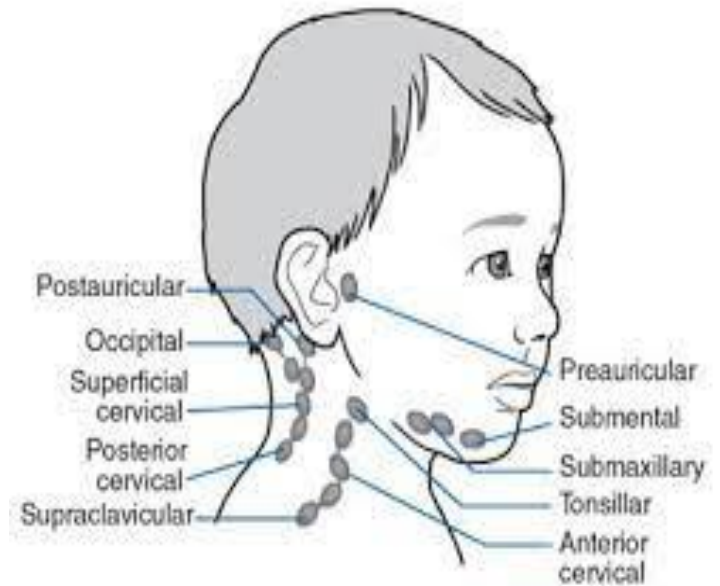
- GA: apparent state of health; level of consciousness; signs of distress; weight/Height; skin color; facial expression; odor of the body and breath; posture, or gait



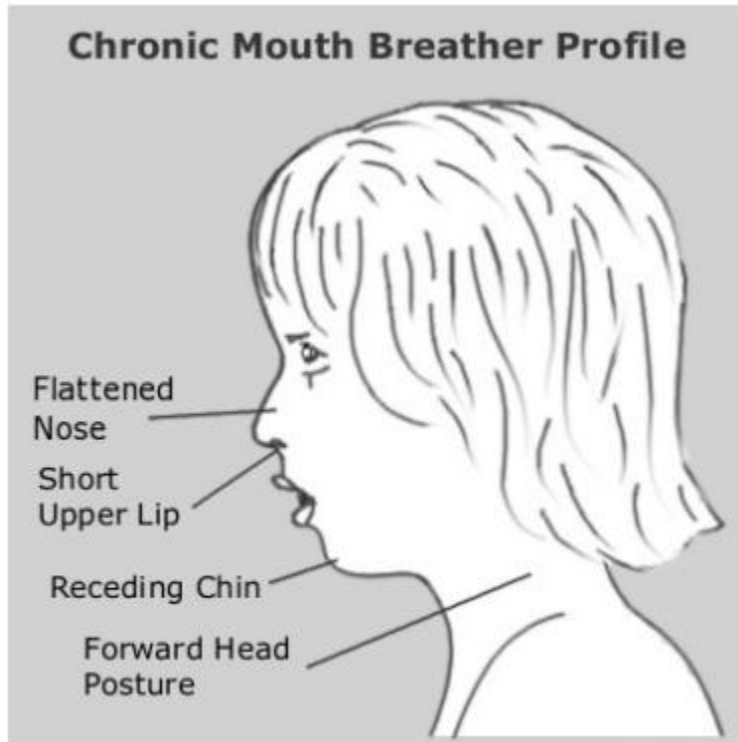
*-TMJ: clicking or
Crepitating sounds and
Range of motion
Normal=35-50mm*

EOE

Facial symmetry



Lips and muscles of mastication



IOE

➤ What are we looking for inside the oral cavity?

- 1) Caries lesion
- 2) Gingiva and periodontium
- 3) Dental trauma
- 4) Dento-alveolar cyst
- 5) Alveolar ridge
- 6) Mal-occlusion
- 7) Prospective abutment

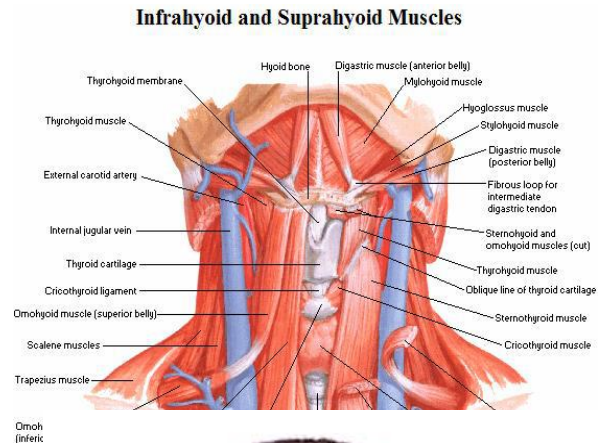
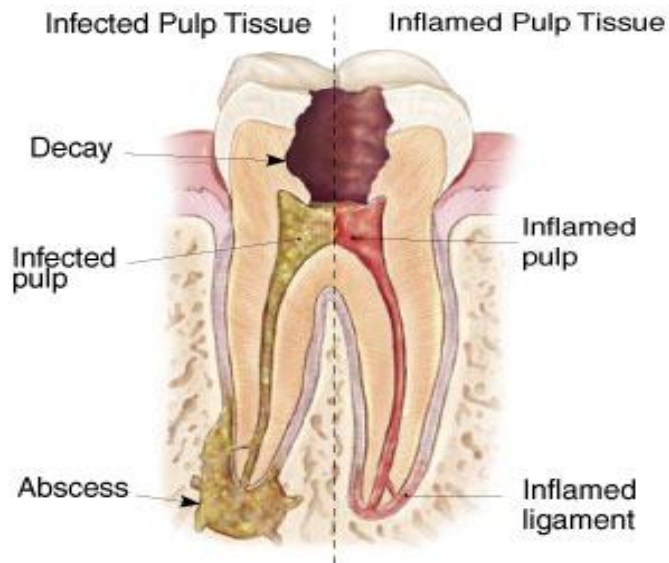
1. Presence of caries lesion



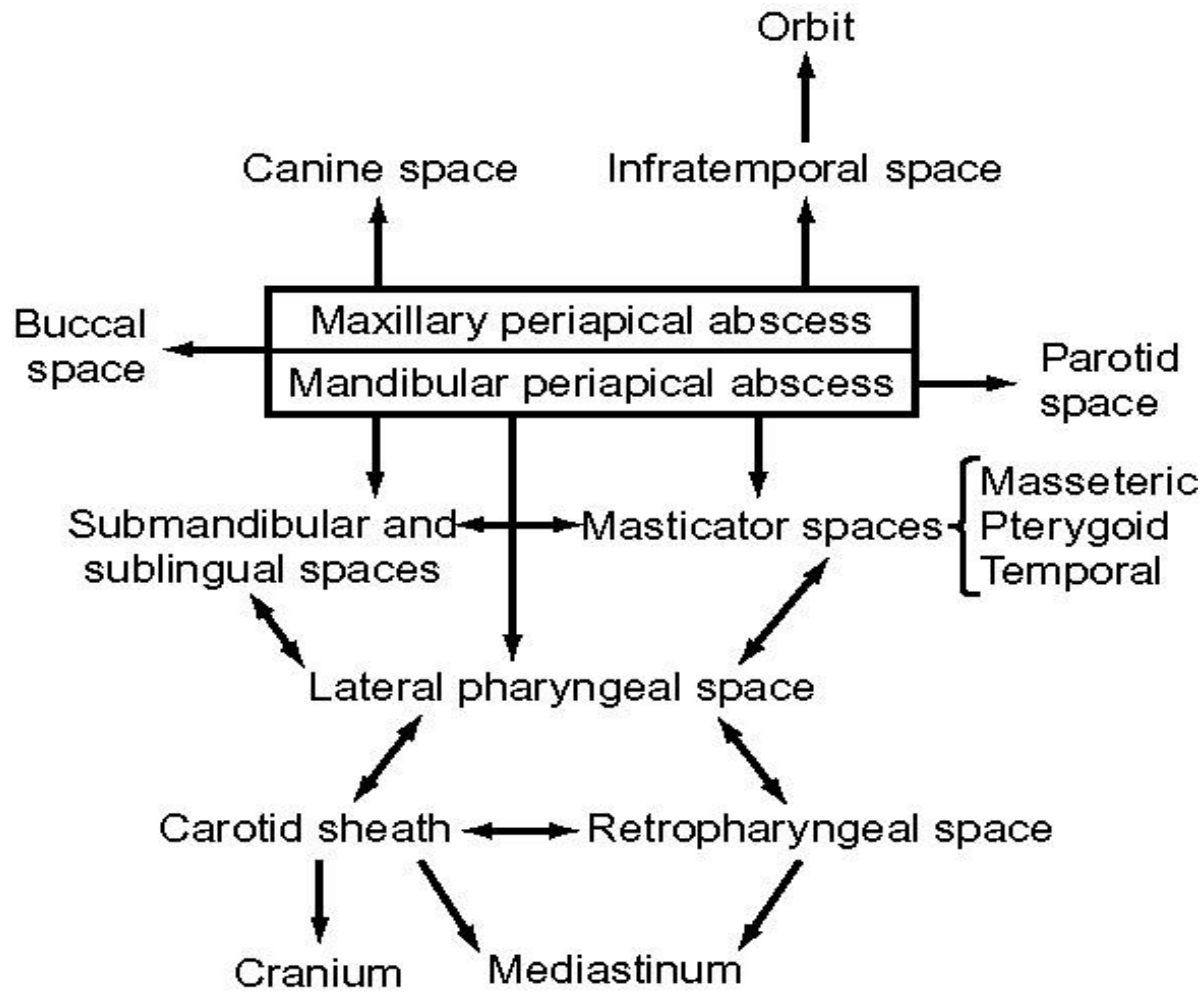
THE STAGES OF CARIES DEVELOPMENT



Primary Complications of dental caries



Secondary complications



2. Gingival and periodontal diseases



Calculus and plaque deposition



Healthy versus unhealthy periodontium



(From Haring JJ, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)
Fig. 33-1 Healthy alveolar crest, normal lamina dura, and periodontal ligament space on a periapical radiograph.

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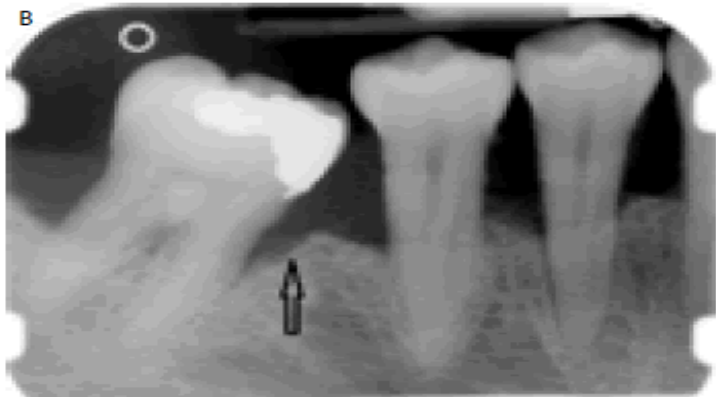


Figure 5b: Mandibular right molar periapical radiograph showing one-wall defect-interdental bone slopes down from facial lingual wall crest of bone towards the crest of destroyed facial lingual bone.

Abscesses in peri-odontium

Periapical



It forms at the root tip.

Gingival



It forms in the space between the gum and tooth.

Periodontal



It forms in a periodontal pocket.

Pericoronal

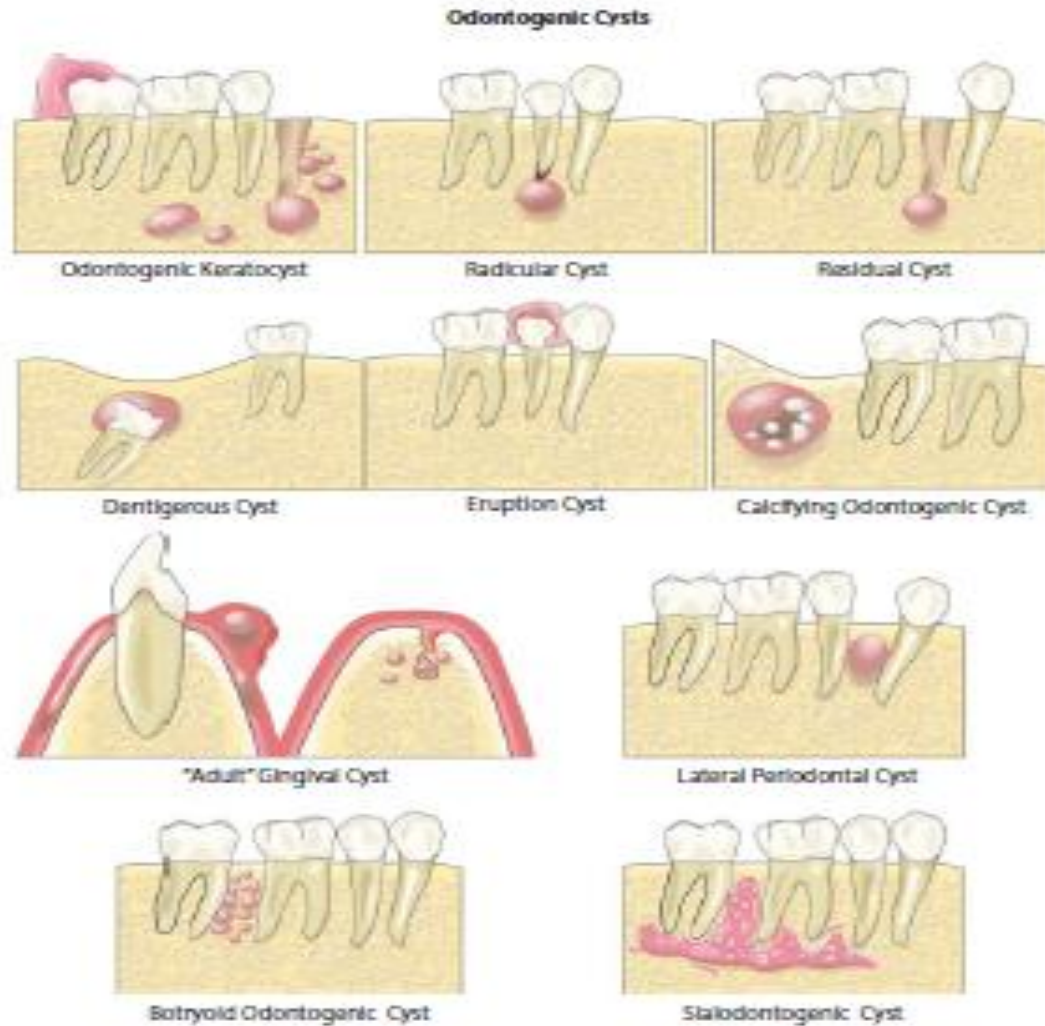


It forms around impacted or partially erupted tooth.

3. Dental Trauma



4- Dento-alveolar cysts



Dentigerous versus radicular cysts



5. Alveolar ridge

- Number and dimensions of edentulous spans should be recorded;
- form, texture, and color of ridge mucosa should also be examined



6:Occlusion



- Type of occlusion (CR, CO, MI, initial contact)
- Wear facets (localized or widespread).
- Presence of any premature contacts.
- Lateral and protrusive contacts (Cuspal interferences in eccentric movements).

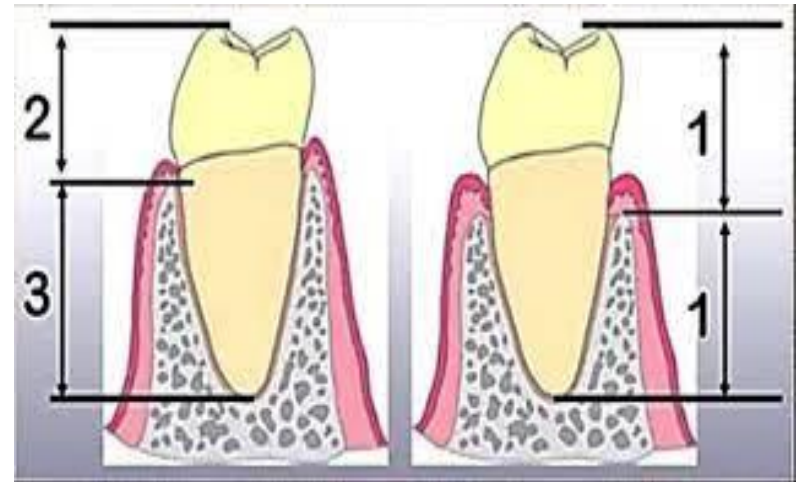
Indications for RPD in the presence of abutment teeth

➤ **RPD or FPD?**

- Edentulous spaces greater than two posterior teeth
- Anterior spaces greater than four incisors
- Spaces that include canine and two contiguous teeth

7- Clinical evaluation of prospective abutment

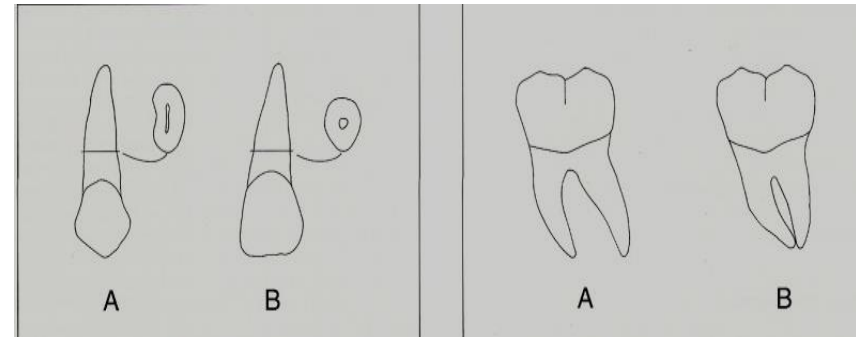
- Crown (color, morphology, mechanical damage, caries lesion, inclination)
- Pulp (vitality)
- Root (configuration, periodontium, size)
- Crown to root ratio (C:R ratio at least 1:1)



A clinical guide in bridge design

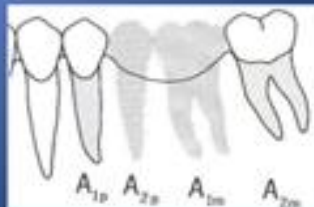
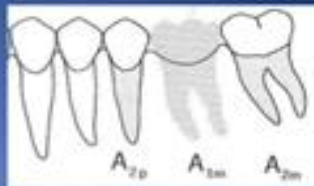
Ante's Law

The Combined pericemental area of all abutment teeth supporting a fixed dental prosthesis should be equal to or greater in pericemental area than the tooth or teeth to be replaced- GPT



Ante's Law states:

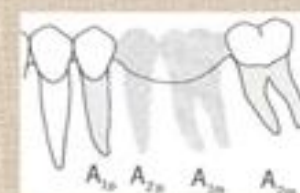
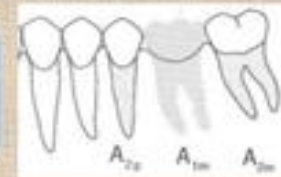
The total root surface area of all supporting teeth must equal or exceed the total root surface area of the teeth being replaced.



Periodontal ligament area

"Ante's Law"

The root surface area of the abutment teeth had to equal or surpass that of the teeth being replaced with pontics.



Why Ante`s law?

- To prevent masticator force overload
- The force generated during routine mastication of food such as carrots or meat measured in males molar region is 39 kg (382 N) and 18 kg (176 N) in the incisor region. The corresponding values for the females is 22 kg (216 N) and 11 kg (108 N) respectively.

Cont`d

❖ Factors modifying Ant's law:

- Well motivated patients and highly proficient in plaque control.
- Opposing occlusion (sound natural dentition or artificial removable prosthesis.
- Decrease in mesio-distal length of edentulous span due to bodily movement of teeth.

Special problems with regards to abutment

Problems	Meaning	Management
<i>Pier abutments</i>	Edentulous spaces that occur on both sides of a tooth, creating freestanding pier abutment.	The use of non-rigid connector on stress breaking spot
<i>Tilted molar abutments</i>		
<i>Canine replacement fixed partial denture</i>	FPDs replacing canines can be difficult because the canine often lies outside the inter abutment axis.	Implant restoration replacing the canine should be considered.
<i>Cantilever fixed partial dentures</i>	FPDs in which only one side of the pontic is attached to a retainer are referred to as cantilevered.	Implant restoration and RPD should be considered.

Diagnostic cast analysis

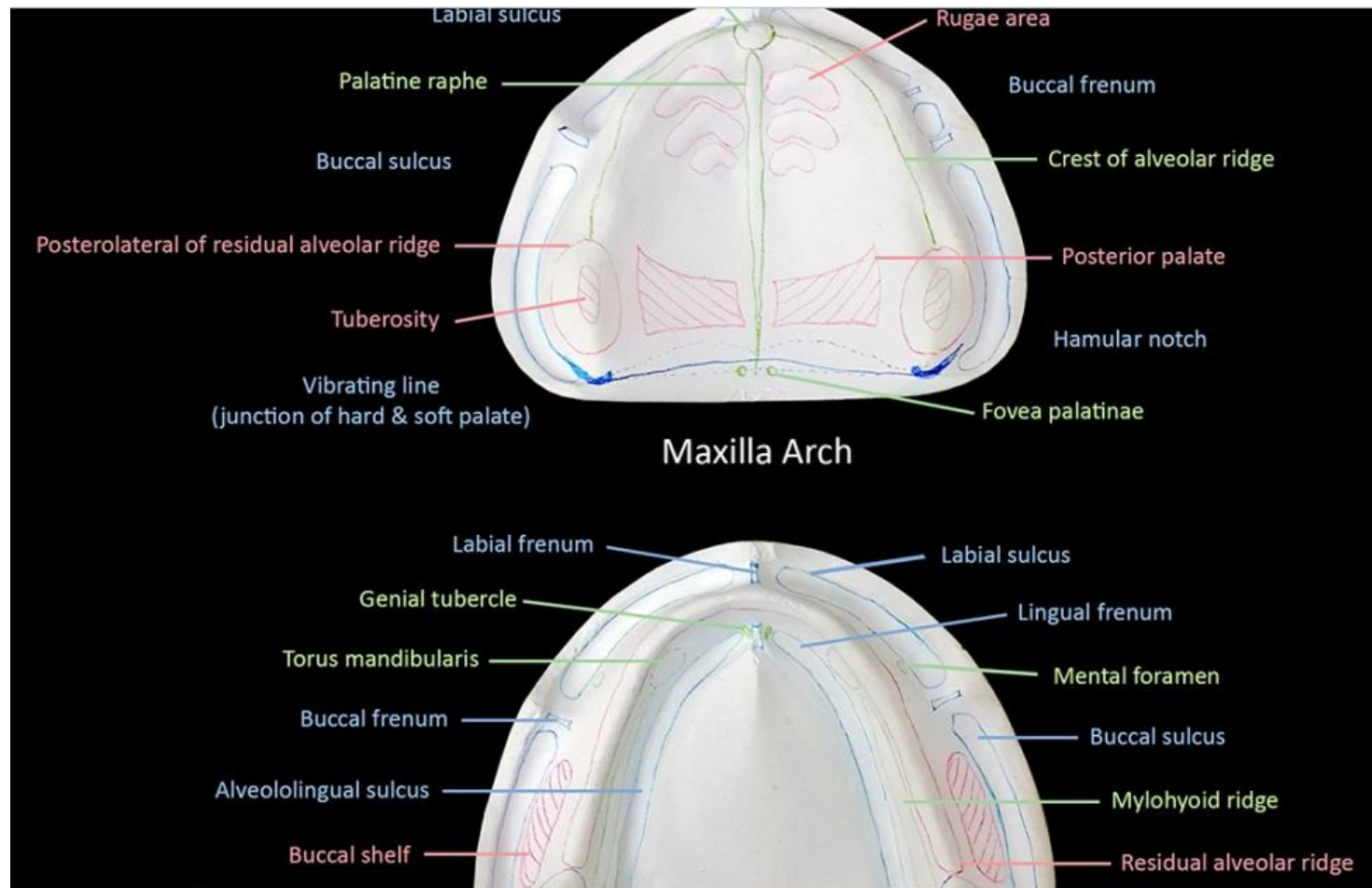
Purposes are to examine:

- relevant maxillary and mandibular land marks
- Occlusion discrepancies
- Length of edentulous span
- the abutment teeth, etc.

➤ **Criteria of good diagnostic casts:**

1. Accurate reproduction of both arches.
2. No bubble or nodules on occlusal surfaces.
3. Can be mounted in CR on a semi-adjustable articulator by means of a face-bow and occlusal wax records.

Important anatomical land marks on maxilla and mandible



Radiographic evaluation





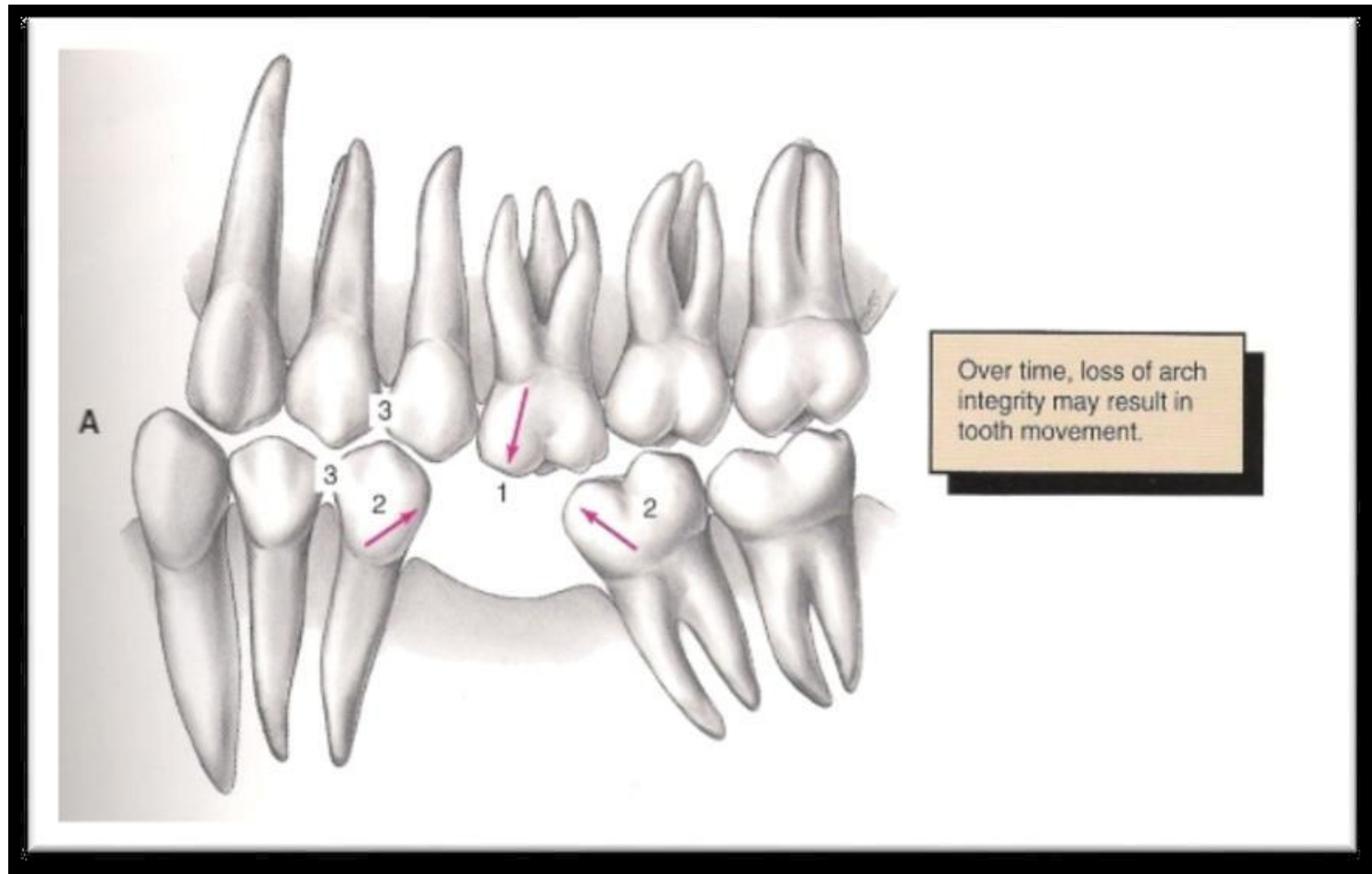
Treatment planning

- Treatment planning is formulating a logical sequence of treatment designed to restore the patient's compliant in a way it would prevent future complication.
- What does the patient needs?
 - 1) Correction of existing disease?
 - 2) Restoration of function?
 - 3) Improvement of appearance?
 - 4) Fixation of social discrimination (bad breath?)

Cont`d

- 1) Treatment planning for single tooth restoration:
Glass ionomer; Composite resin; metal inlay ;
Ceramic inlay or MOD Onlay.
- 2) Extra-coronal restorations: Partial veneer crown; full
metal crown; metal ceramic crown; all ceramic
crown; resin veneer; ceramic veneer; fiber-
Reinforced Resin.

Does removal of a tooth is part of treatment planning?

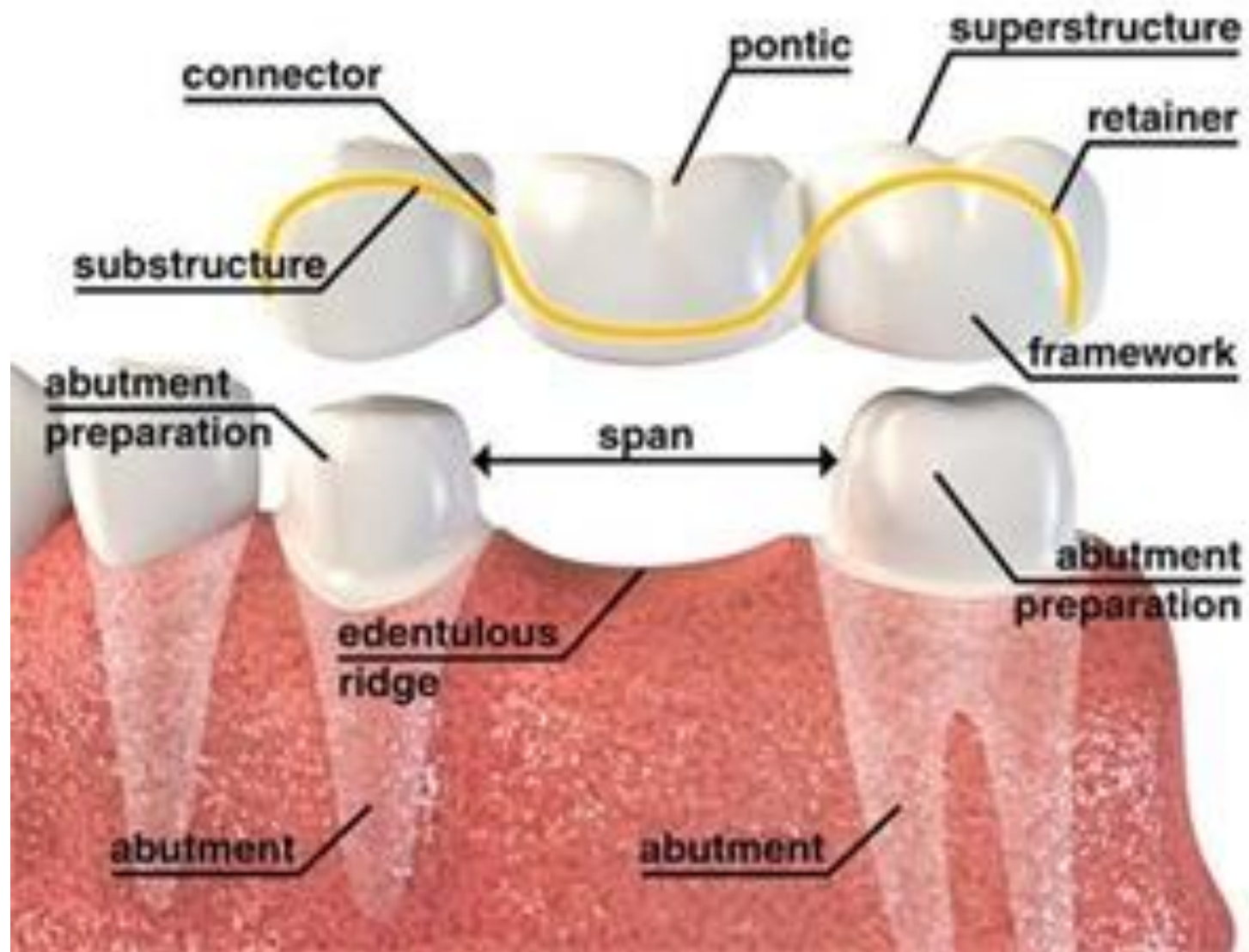


Treatment planning for the replacement of missing teeth

- 1) Removable partial denture.
- 2) Tooth-supported fixed partial denture.
 - Conventional fixed prosthesis.
 - Resin-bonded fixed prosthesis.
- 3) Implant supported fixed prosthesis.
- 4) No prosthetic treatment.

Replacement of missed tooth





Frame

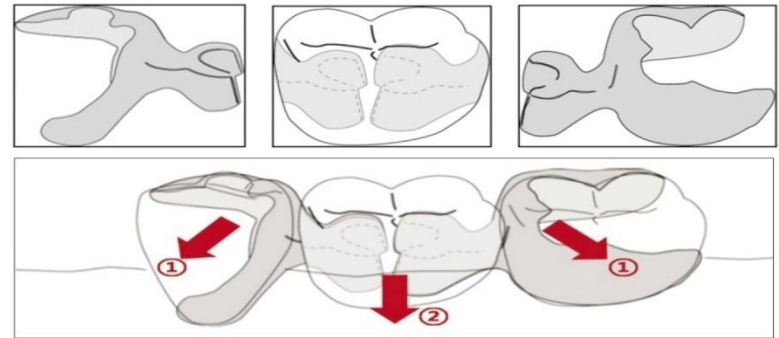


Indications of RPD

- 1) Edentulous spaces greater than two posterior teeth
- 2) anterior spaces greater than four incisors.
- 3) Spaces that include a canine and two other contiguous teeth.
- 4) An edentulous space with no distal abutment.
- 5) Multiple edentulous spaces.
- 6) Bilateral edentulous spaces with more than two missing teeth on one side.
- 7) Periodontally weak primary abutments.
- 8) An insufficient number of abutments for FPD.

Cont`d

- Resin-bonded fixed prosthesis (RBFPDs) are minimally invasive, requiring less tooth preparation compared to conventional FPD.
- They are usually indicated for short spans of two or three units.



Cont`d

- Conventional fixed prosthesis: Conventional fixed bridgework involves preparing one or both teeth adjacent to the gap to receive a crown.
- Conventional bridges are more invasive as they require reduction of the supporting abutment teeth which results in irreversible loss of tooth enamel.



Cont`d

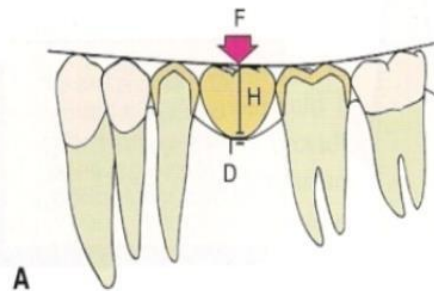
- If a patient presents with a long-standing edentulous space into which there has been little or no drifting or elongation of the adjacent or opposing teeth and there is no functional occlusal and esthetic impairment; **NO PROSTHOSIS IS REQUIRED.**

Factors affecting the selection of prosthesis type

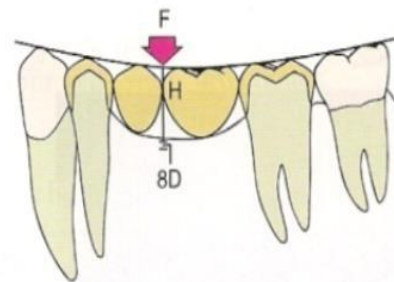
- 1) The biomechanical considerations (distribution; **length** and arch form)
- 2) The prospective abutment.
- 3) Esthetic requirements.
- 4) Patient's desires.
- 5) Financial factors.
- 6) Clinician's skills.
- 7) Laboratory support.
- 8) Patient's motivation and expected cooperation.

Length

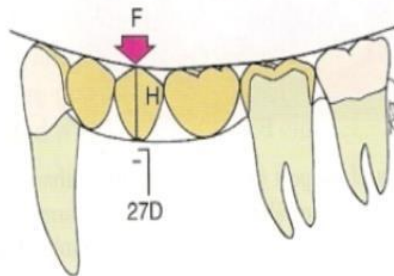
- The longer the span the greater the flexing.
- Bending or deflection varies directly with the cube of the length and inversely with the cube of the occluso-gingival thickness of the pontic.
- Compared with a FPD having a single tooth pontic span, a two-tooth pontic span will bend 8 times as much. A three-tooth pontic will bend 27 times as much as a single pontic.



A



B



C

Excessive span length is a common contraindication for fixed dental prostheses.

Fig. 3-26

The deflection of a fixed dental prosthesis is proportional to the cube of the length of its span. **A**, A single pontic deflects a small amount (D) when subjected to a certain force (F). **B**, Two pontics deflect 2^3 times as much ($8D$) to the same force. **C**, Three pontics deflect 3^3 times as much ($27D$).

Treatment sequence

1. Treatment of symptoms

- Discomfort.
- Pain.
- Swelling.
- Sensitivity.

2. Stabilization of deteriorating conditions.

Dental caries.
Periodontal disease.

3. Definitive therapy.

Oral surgery.
Periodontics
Endodontics.
Orthodontics
Prosthodontics.

- Occlusal adjustment.
- Anterior restorations.
- Posterior restorations.
- Complex prosthodontics.

4. Follow up and maintenance.

Clinical steps of fixed prosthesis construction

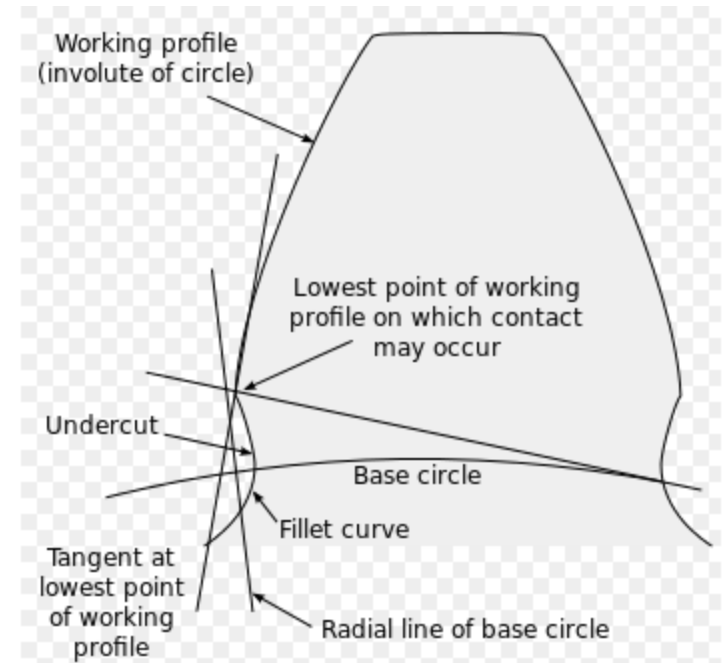
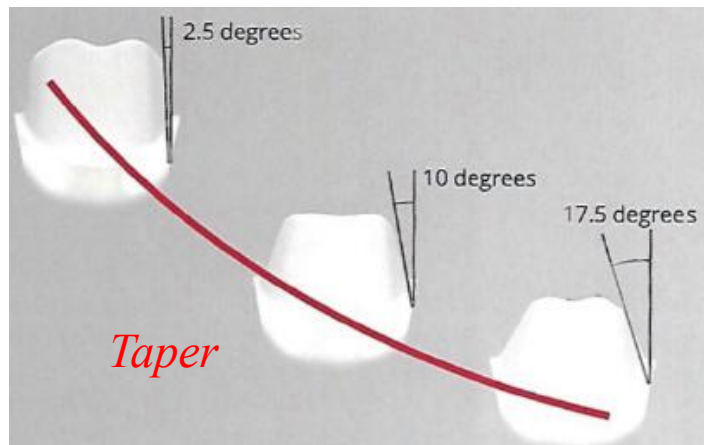
- History, P/E and selection of prospective abutment
- Treatment planning for fixed prosthodontics
- Shade selection
- Tooth preparation
- Impression taking
- Prosthodontic request prescription
- Cementation
- Occlusion verification
- Complications following fixed prosthodontics

Basic principles of tooth preparation

➤ Discuss in a group of four students:

- 1) The basic principles of tooth preparation?
- 2) Terminologies like (undercut, taper, retention, stability and resistance)

Cont`d



Tooth preparation is the mechanical alteration of a defective ,injured or diseased tooth to receive a restorative material that re-establishes a healthy state for the tooth which needs to meet certain requirements and standards.



General agreements

- The general shape of the preparation should follow the anatomical contours of the natural tooth with smooth and rounded transitional line angles.
- Occlusal reduction should also follow the existing anatomy, including pits and fissures, and reproduce the main ridges, allowing an appropriate occlusal morphology of the restoration.
- The teeth should be uniformly reduced to ensure a restoration with suitable shape and esthetics.
- The path of insertion of a preparation for a ceramic crown should parallel the long axis of the tooth.

Cont`d

Mechanical principles

Retention

Stability

Resistance

Biological principles

Preservation of tooth structure

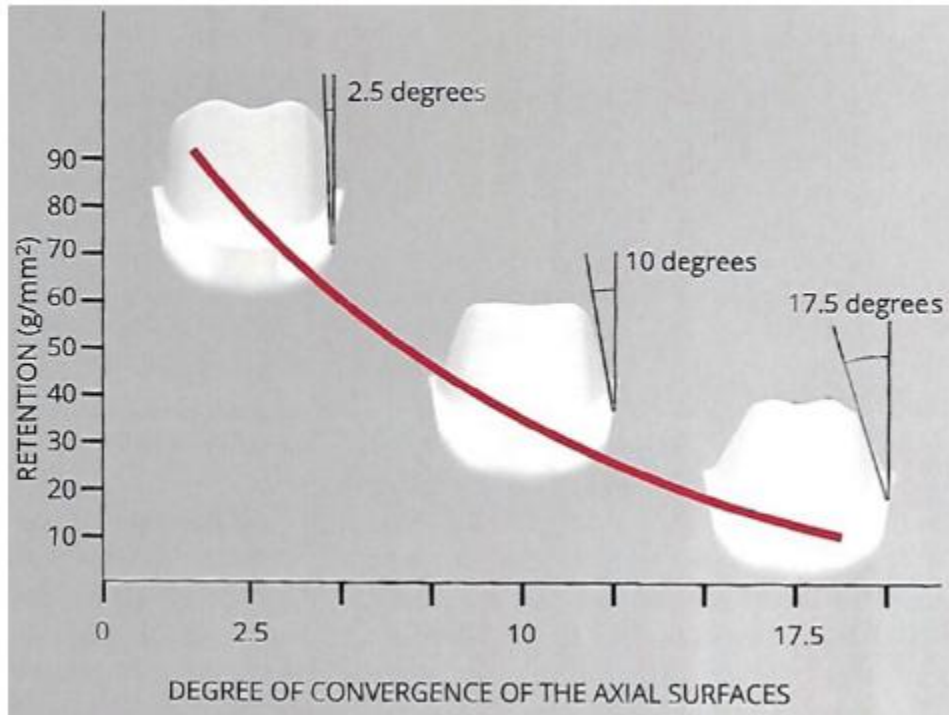
Preservation of periodontium

Esthetic principles

Retention

- It is the ability of the restoration to resist a displacement along the long axis of the tooth preparation in coronal direction.
- This basically depends on the frictional resistance that is the existing contact between the internal surfaces of the restoration and the external surface of the prepared tooth.
- For the quality of the luting cement alone would not be enough for retention the geometric configuration of the tooth preparation must place the cement in compression to provide the necessary retention.
- These includes : taper; length/surface area; and path of insertion

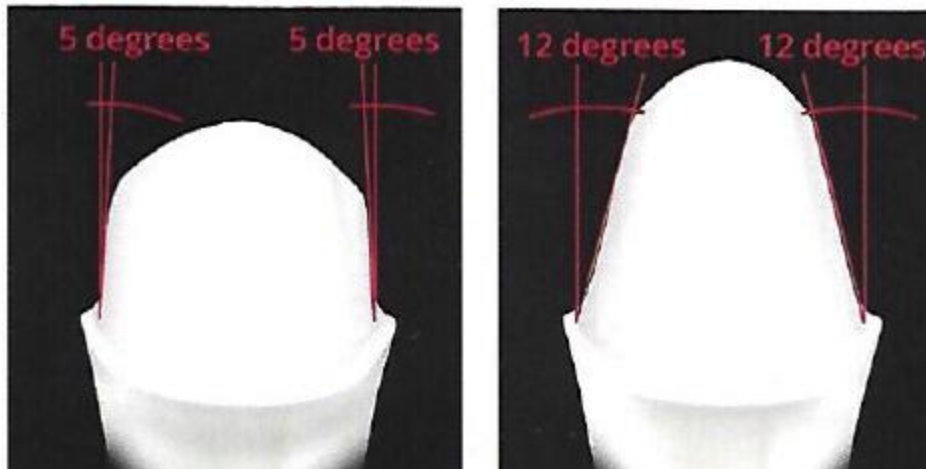
*Taper: it is the degree of convergence of tooth preparation
Ideally must be 2-4 degree but still 4-22 degrees are acceptable*



Preparation walls are tapered to allow their visualization, prevent undercuts, compensate for inaccuracies in the fabrication process, and permit more nearly complete seating of restorations during cementation.

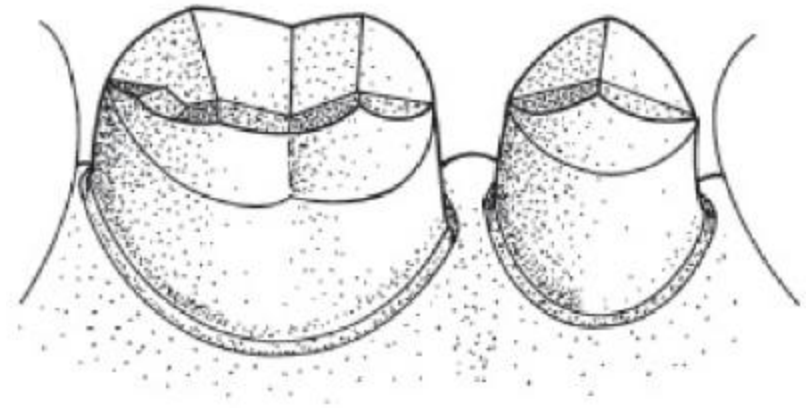
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- The degree of convergence will be determined by the individual characteristics of each preparation.
- Teeth with short clinical crowns require less convergence to provide greater retention, and teeth with long clinical crowns require greater convergence.

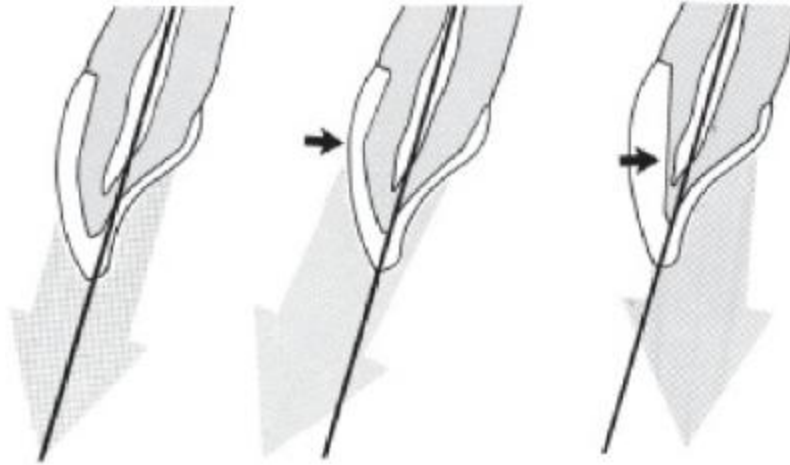
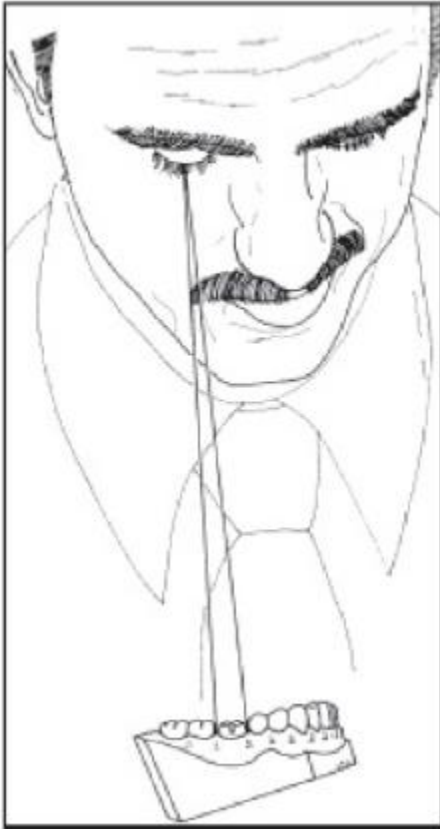


Length and surface area

- Longer preparations will have more surface area and therefore will be more retentive.
- The greater the clinical crown of a prepared tooth, the larger the contact surface and the greater the final retention.



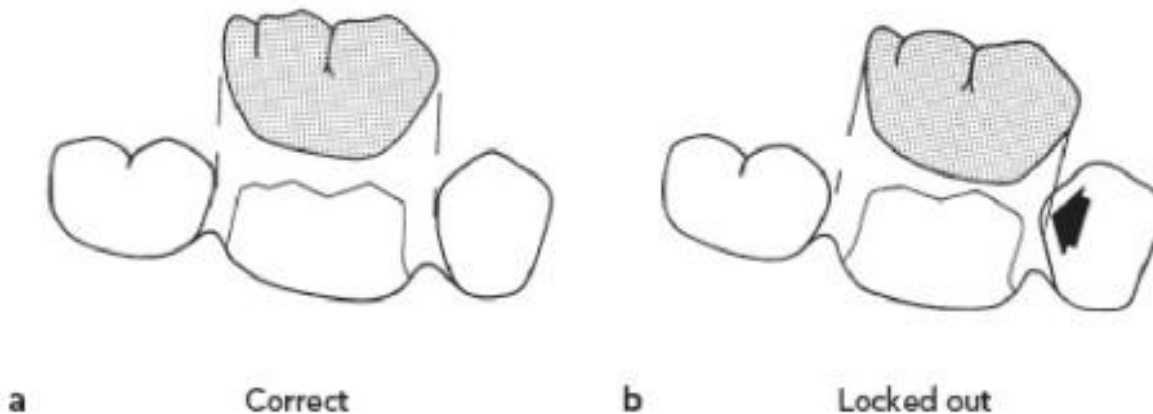
Path of insertion: The path of insertion is an imaginary line along which the restoration will be placed onto or removed from the preparation



The path of insertion of a preparation for a ceramic crown should parallel the long axis of the tooth.

Cont`d

- The mesio-distal inclination of the path must parallel the contact areas of adjacent teeth. If the path is inclined mesially or distally, the restoration will be held up at the proximal contact areas (ie, locked out).

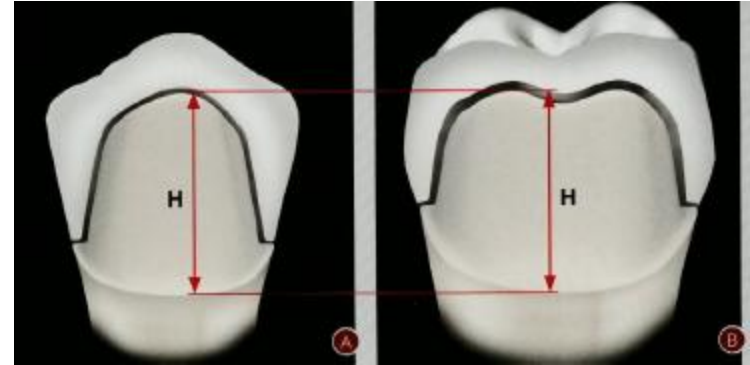


Stability

- Stability is the property of the preparation to withstand the displacement of a restoration due to oblique forces.
- These can lead to rotation of the restoration such as during mastication or in the presence of para-functional habits.
- Preparations with a high degree of convergence or short preparations can be subjected to dislodging forces in various directions.
- To obtain stability in preparation, two alternatives can be employed: decreased convergence of surfaces and adding grooves to occlusal surfaces.

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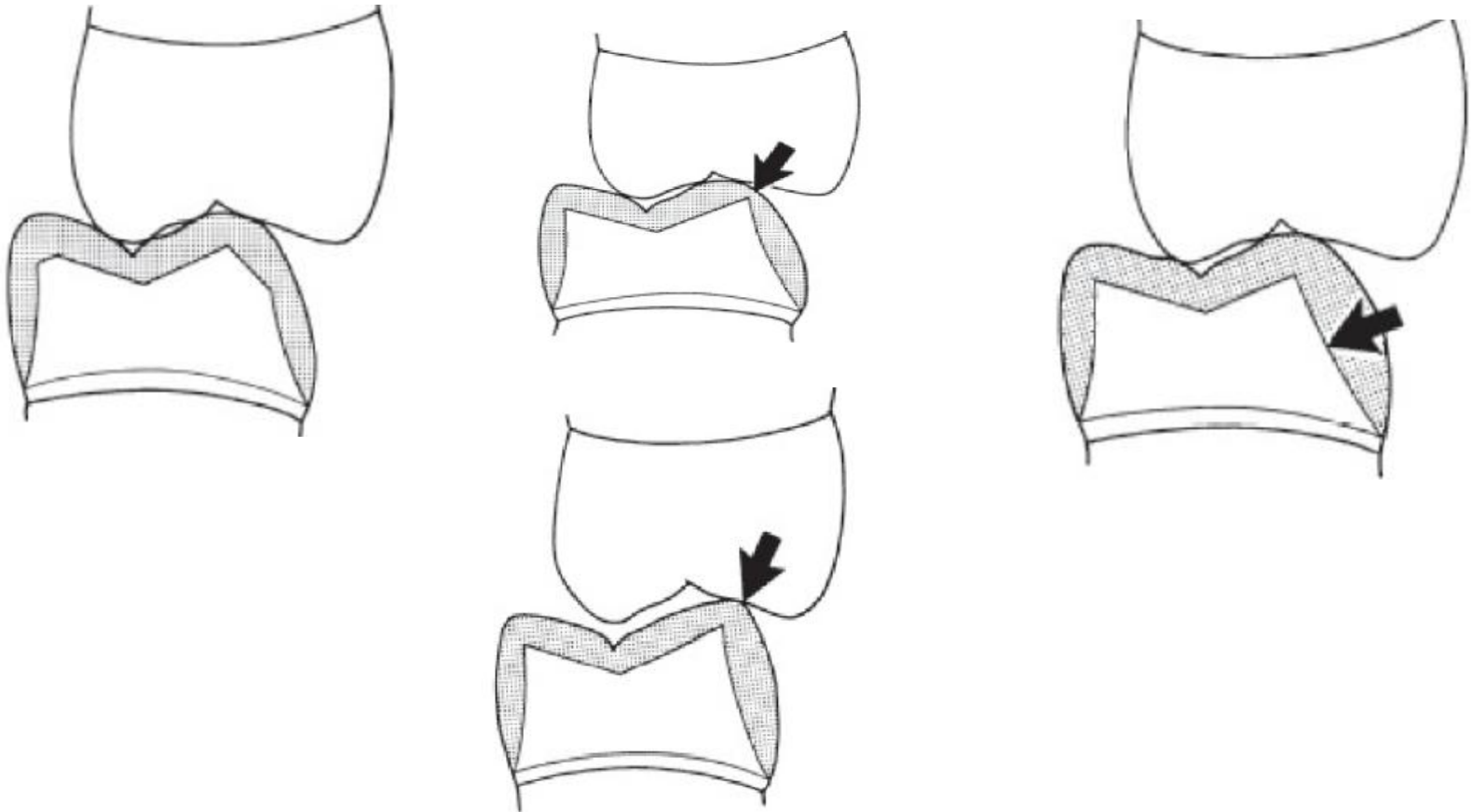
- The length of the tooth should always be greater than its base to ensure stability.
- Based on the principles of the height/width ratio if the width is greater than the height, the radius of rotation increases, impairing adequate stability.
- For the same height, crowns with larger diameters have comparatively less stability.



Resistance

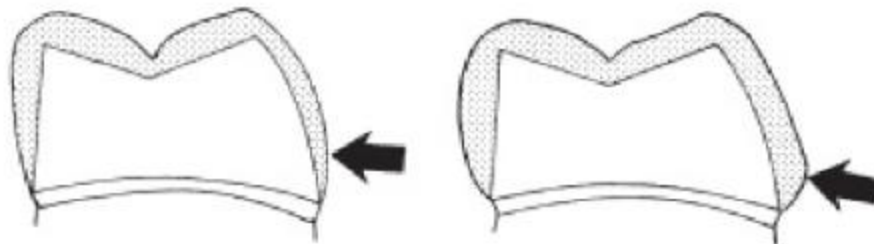
- The structural stability is the minimum material thickness of the restoration to resist the action of masticator forces without deformation.
- The recommended thickness of the material may vary from one region to another but for all-ceramic restorations, there must be enough space to accommodate 1.5 to 2 mm of the ceramic material.
- One of the most important features for providing adequate ceramic bulk and strength to the restoration is occlusal clearance.
- There should be at least 2mm of clearance on preparations for all-ceramic crowns.

An integral part of the occlusal reduction is the functional cusp bevel (which has to be wide)

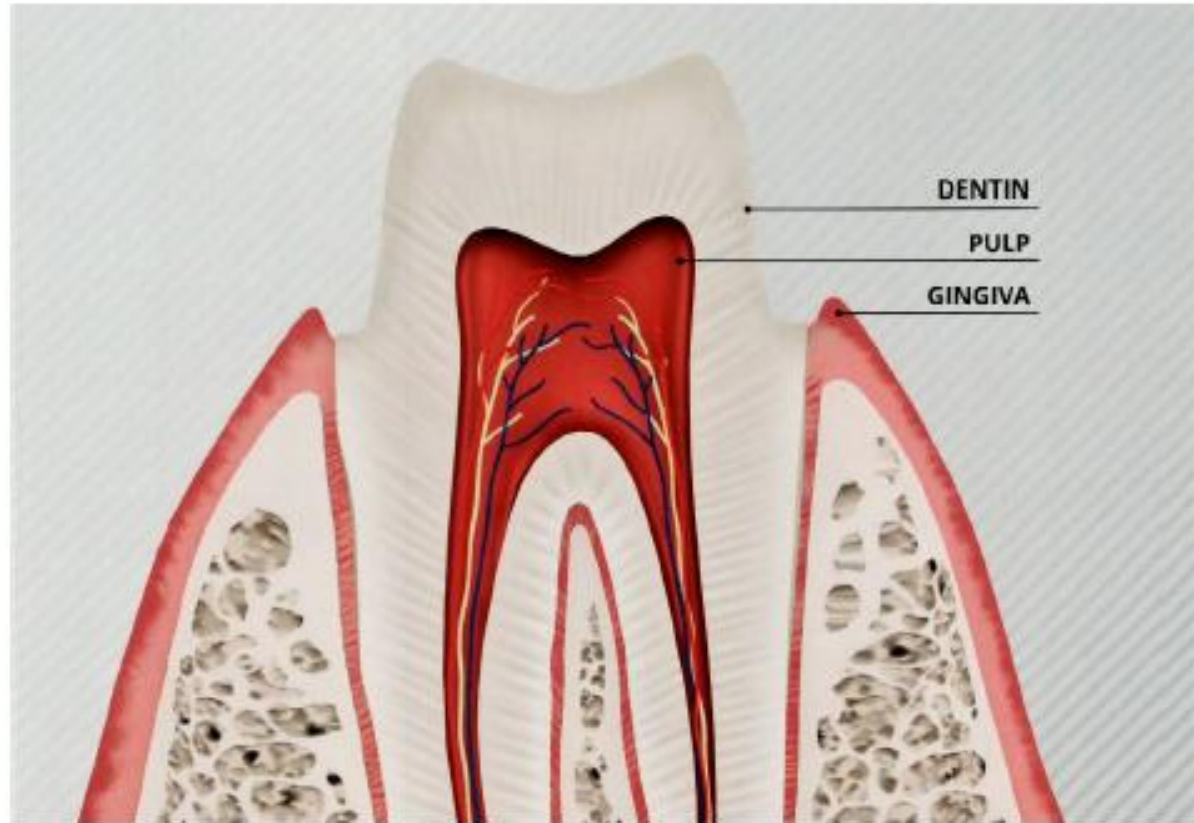


Axial wall restoration

- Inadequate axial reduction can cause thin walls /a weak restoration or over contoured restoration
- The angles of the preparations should be rounded to increase the strength of the ceramic restorations and to facilitate laboratory manufacturing steps. Acute angles on the prepared surfaces act as stress concentration regions.



Biological principles (preservation of the tooth and periodontium)



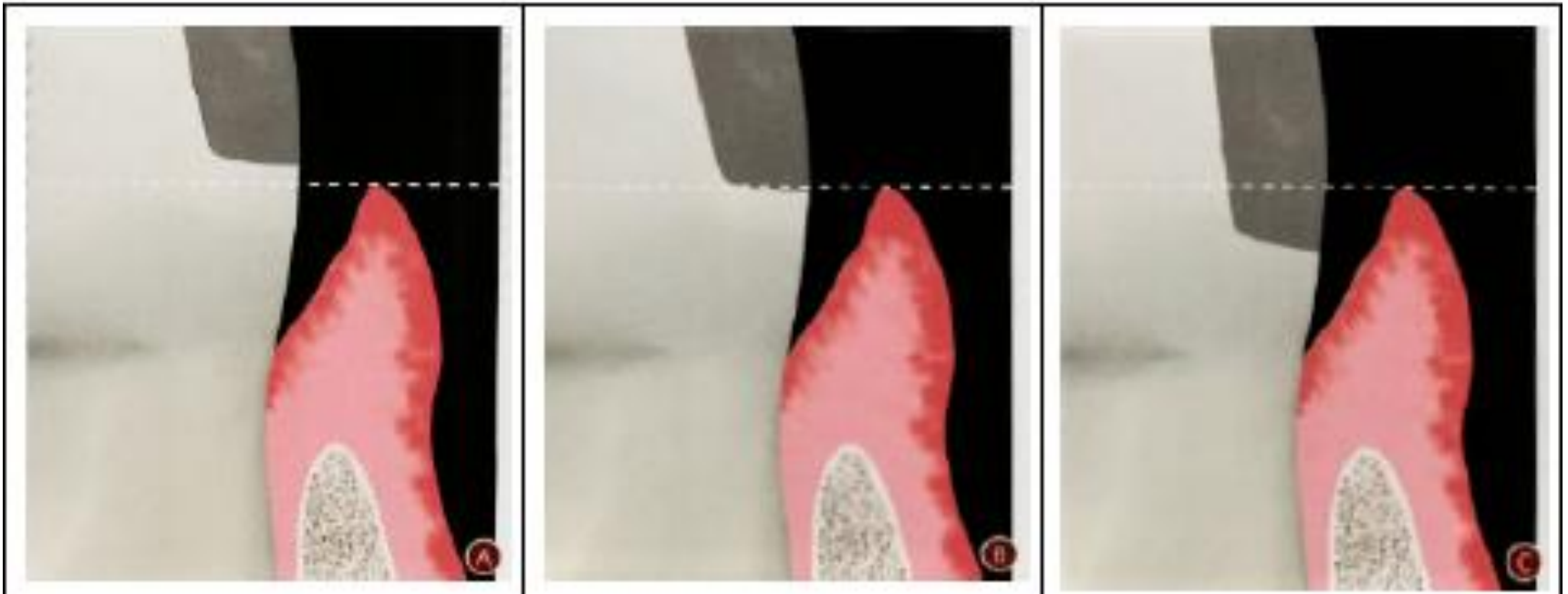
Would the pulp react to tooth preparation???

- A temperature increase of more than 5°C in the dental pulp during tooth preparation can result in a 15% incidence of necrosis.
- Example worn-out diamond burs require more pressure from the clinician which generates more heat on the tooth surface which is adequate enough to destroy the pulp.
- Preparation at high speed is less harmful to the pulp than preparation at low speed due to the decrease of vibration and heat generation.
- Preservation of 1 to 2 mm of dentin during tooth preparation is sufficient to prevent damage to the pulp.

The periodontium forms the basis for esthetics, function, and comfort of the dentition.

- To control biofilm, the restoration margins of the preparation must have regular ends and be positioned in areas with better access.
- It is essential that the edges are sharp and well defined in all indirect restorations.
- Ideally, the end should be maintained as far as possible from the gingiva (supra-gingival), because deeper locations of the restoration margins into the gingival sulcus result in more inflammatory responses which would be main risk factors for periodontitis.

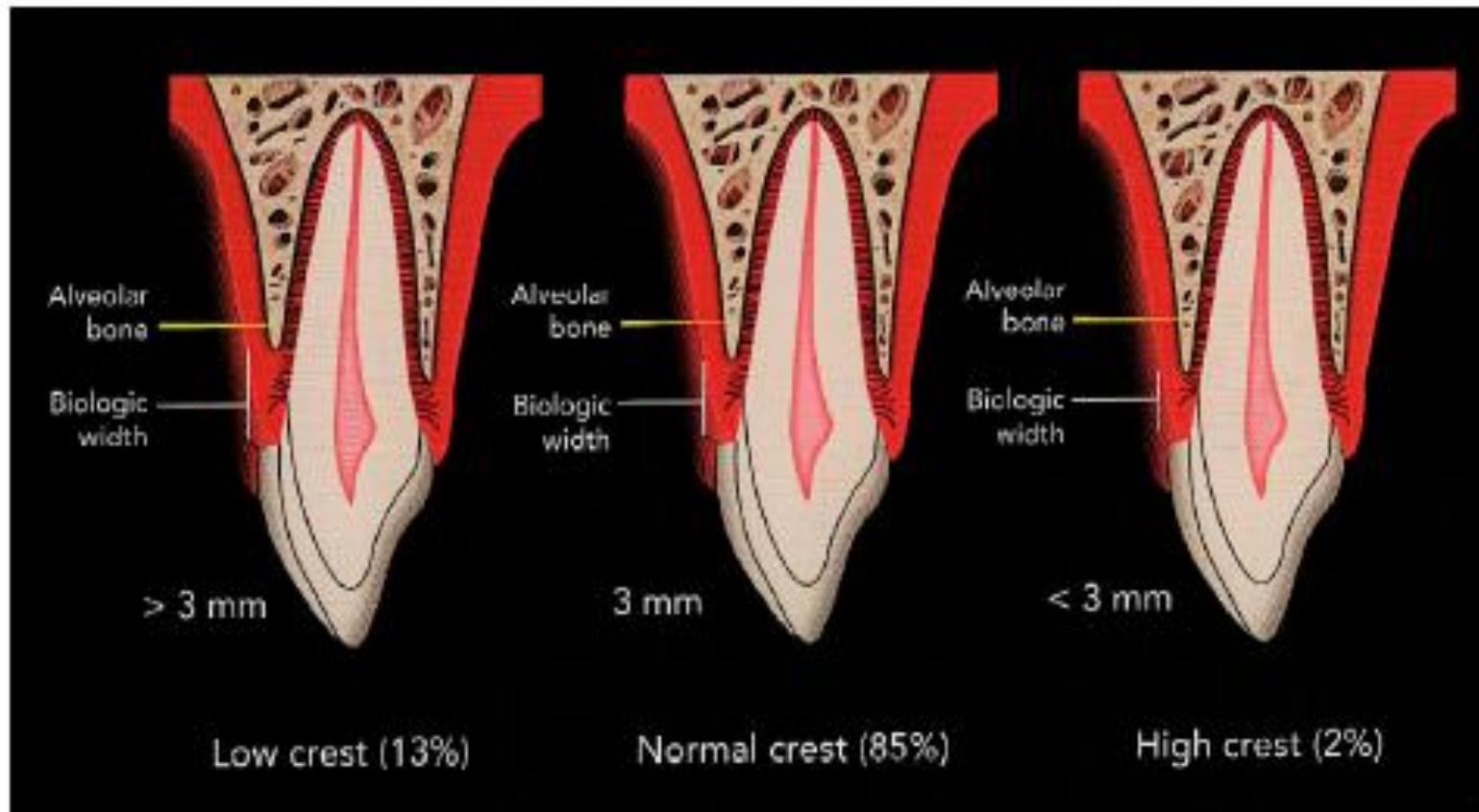
Although the restoration margin should preferably be placed supra-gingivally (A), for esthetic reasons it is sometimes placed gingivally/equigingivally (B) or subgingivally (C).



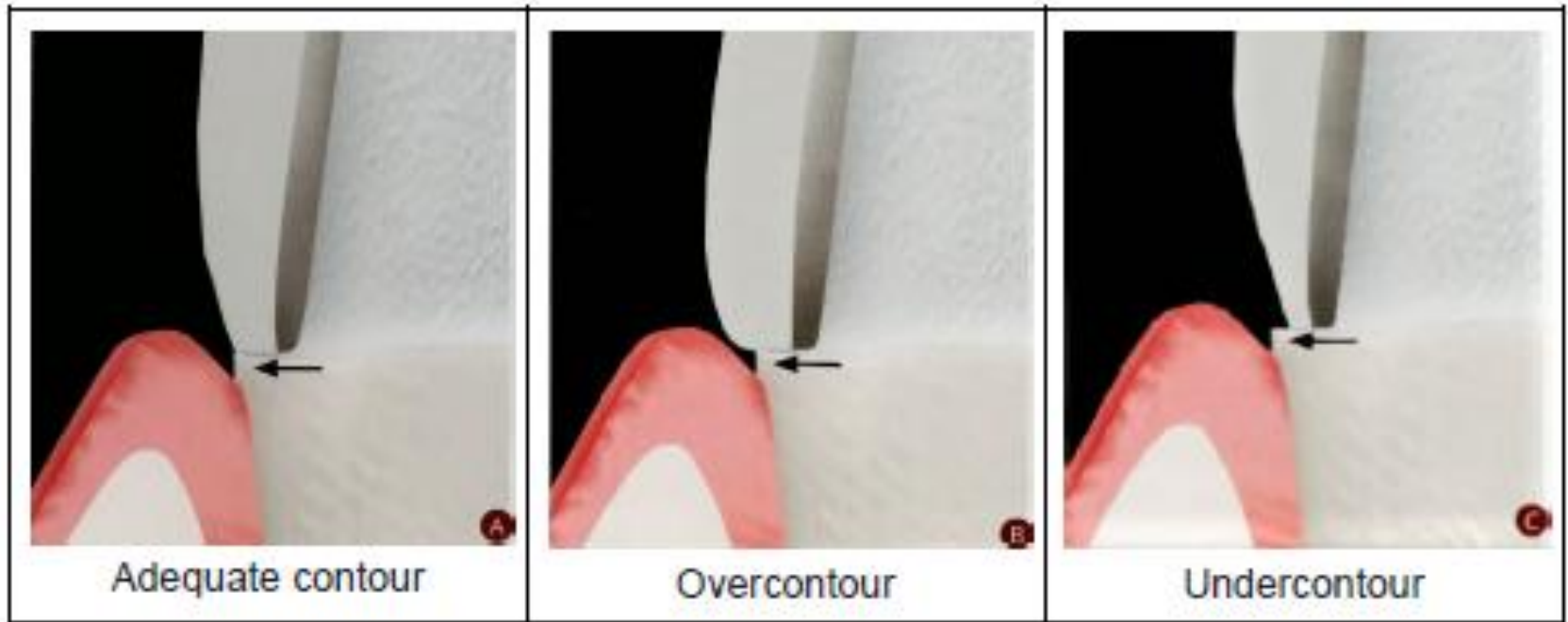
Retraction cord



Prior assessment of crest position to use retraction cords without complication



Adequate Contouring



Esthetic principles

- The composition of a smile that is considered beautiful, healthy and attractive involves a balance between the shape and symmetry of the teeth, lips and gums, including the way in which these elements relate to and harmonize with the patient's face.
- Importantly, all of these references should be sought in conjunction with functional principles of occlusion, incisal guidance, canine guidance, overbite, and overjet.

Before



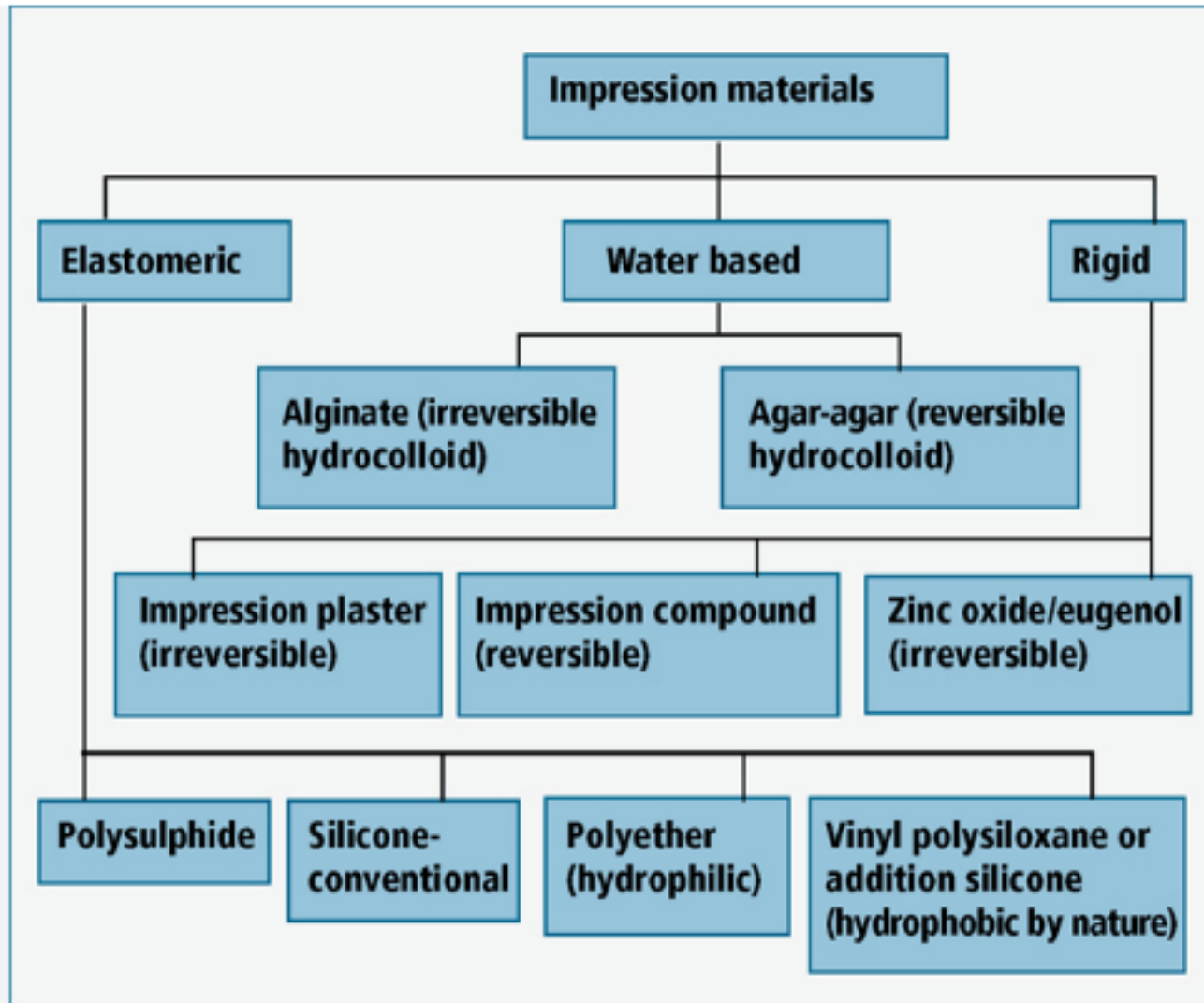
After



Impression taking



Materials



Cont`d



Tray selection

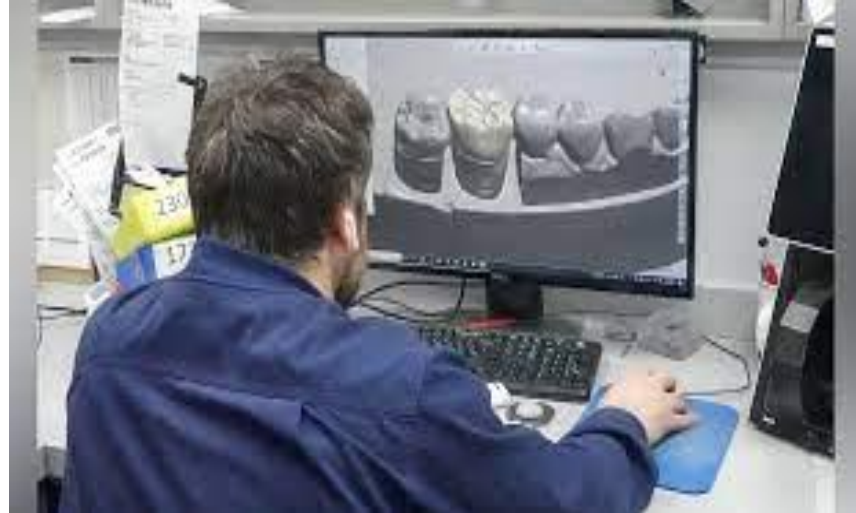


TradeKey.com

Qualities of good impression

Criteria	Description of Error
Finish Line, Void/Bubble	Any detectable void on the cervical finish line of a preparation.
Finish Line, Lack of Wash Material	Cervical finish line recorded in heavy body or putty material with no wash above or below the finish line. Monophase excluded from error.
Tray, Inadequate Retention of Material	Impression material pulling away from tray or not engaging tray retention features.
Tray, Pressure of Tray On Soft Tissue	Vertical tray flanges exposed by displacement of impression material. Any occurrence within 2 teeth of preparation(s) or on the preparation(s).
Tray, Show Through of Occlusal/Incisal Edges	Horizontal tray areas visible by displacement of impression material. Any occurrence within 2 teeth of preparation(s) or on the preparation(s).
Material, Inadequate Fusion of Viscosity	Lack of complete fusion between body and wash materials.
Material, Void on Preparation	Voids not located on the finish line greater than 1 mm in size
Material, Lack of Polymerization	Impression material visibly unset or tacky to the touch.
Gingival Displacement, Tissue Over Finish Line	Lack of flash beyond the cervical finish line, detected by change of reflection or visible horizontal bur marks on the preparation for ill-defined margins.
Gingival Displacement, Blood On Impression	Blood, coagulant, or any foreign materials around the cervical finish line.
Dual Arch, Lack of MIP	No thinning of impression material over occlusal contacts. Detected by holding impression against light source.
Dual Arch, Unprepared Stops	Lack of unprepared teeth anterior and posterior to the preparation(s).
Dual Arch, Canine Recorded	Lack of registering the complete maxillary and mandibular canine teeth.

Alternative to impression taking (CAD/CAM)



Shade selection



Recommended protocol for shade selection

1. Remove bright color from the working field. If the patient is wearing bright clothing, it is prudent to cover the patient with neutral color bib (grey). Any dark color lipstick should be removed, because it could affect shade matching
2. Always clean the tooth by using prophylaxis paste prior to shade selection
3. Its important not to view the shade comparision for more than 7 s to avoid eye fatigness
4. Clinician should be at a distance of 28-33 cm from the patient during shade selection
5. Always determine shade when the teeth are most hydrated, because enamel dehydration reduces its translucency by 82%, misleading the clinician
6. Shade comparision should always be done in between 10 am and 2 pm, because at this time color tempreature is around 5500 K and then under color corrected light to ensure the accuracy of the match
7. During the shade comparision always place shade tabs either above or below the tooth to be match, never place shade tab adjacent to the tooth to avoid binocular effect
8. Always, value is analyzed first, followed by chroma and then hue
9. Shade selection should not be done immediately after bleaching, patient should be recalled after 2-3 weeks for shade comparision
10. Always during shade selection teeth should be divided in 3 regions. Gingival area (gives accurate determination of dentinal chroma), Body area and Incisal area (enamel is thickest here and varies from translucent to transparent).

Shade guides

- Tooth shade matching is most frequently performed visually using dental shade guide.
- The first shade guide was introduced in 1956 by Vita Zahnfabrik.
- The most popular shade guide is - VITA Classical, Chroma scope, and Vitapan 3D-Master shade guide.

Shade taking devices

Shade taking devices	
Instrument	Type
Shade Eye, Shofu	Colorimeter
Easyshade/Easy Shade	Spectrophotometer
Compact, Vita	
Shade Scan, Cynovad	Digital color imaging/colorimeter
Shade Vision, X-Right	Digital color imaging/colorimeter
Spectro Shade Micro, MHT	Digital color imaging/ spectrophotometer
Clear Match, Smart Technology	Software (requires a digital camera)
Color Scanner 2006, Nuova	Software (requires a digital camera)
Franco Suisse	
DVS11, DV	Spectrophotometer

Communication between dental office and dental laboratory

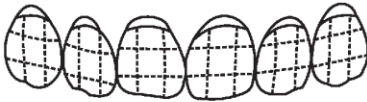
	Dr.
	Date

Patient : Sex : Male / Fem
 Age : Type of Face : ☐ ☐ Shade.....

No. Clinic

Fixed Prosthesis

Shade Instructions



Pontic



*Your choice if not indicated

BRIDGE ☐ UNITS ☐ TEETH NO
 ALLOY : - PRECIOUS ☐ NON PRECIOUS ☐ Zircon Crown ☐
 Fixed Prostho P.F.M ☐ Full Metal ☐ Post & core ☐ Richmond ☐ Implant ☐
 EMPRESS :- Crown ☐ Vencer ☐ Inlay-Onlay ☐ Cosmo Post ☐
 TEXTURE :- SMOOTH ☐ STRTIFLED ☐
 GLAZE :- GLOSSY ☐ LIGHT GLOSSY ☐

Removable Prosthesis

UPPER ☐ LOWER ☐ UPPER & LOWER ☐
 Complete Denture ☐ Partial Denture :- Mefal ☐ Acrylic ☐
 Immediate Denture ☐ Teeth For Extraction.....
 REPAIR ☐ REBASE : SOFT ☐ HARD ☐
 ADDITION ☐ TEETH FOR ADDITION

Reading assignment

➤ What are the special considerations that should be made while preparing tooth for veneer restorations???

Fixed prosthesis cementation and occlusion verification

- Temporary restoration
- Cementation
- Occlusion verification
- Complications following fixed restoration

Discuss in a group of four students:

- 1) Some of the cements used in fixed prosthodontics
- 2) Occlusion (centric relation, centric occlusion, eccentric occlusion, maximum intercuspation, occlusal interferences)
- 3) Common complications following fixed prosthodontics

Are we going to leave the preparation as it is until the permanent restoration comes in?



Temporary dental crowns as a transition

- ✓ Why do we need temporary dental crowns?
- ✓ What are temporary dental crowns made of?
- ✓ How long does temporary dental crowns last?
- ✓ How do you care for temporary dental crowns?



Why provisional restorations?

- For better patient comfort
- Dentin and pulp protection
- To temporarily restore function (mastication, phonetics and aesthetic)
- Space maintenance (prevent drifting of the abutment)
- Assessing pulpal health
- Waiting for gingival healing from periodontal and oral surgery
- Waiting for implants to integrate
- Evaluating aesthetic and functional changes

Cont`d

- Some of the clinical requirements of provisional cements are:
 - 1) Good retention
 - 2) Good marginal seal to prevent micro-leakage and recurrent decay during the provisional phase of treatment
 - 3) Durability, to prevent "wash out" and resulting sensitivity issues, and
 - 4) Ease of cleanup from the inside and outside surfaces

Dental cements



Cementation

- What are common dental cements used for fixed prosthodontics?
- What are the bases to choose an appropriate luting agent for each unique clinical situation?
- What are the steps to be followed for cementations?

Dental cements

- Dental cements are materials that acts as an adhesives to hold together or to join a tooth and to a restorations.
- The American Dental Association and the International Standards Organization (ISO) have teamed up to classify dental cements as

Type I: Luting agents that include permanent and temporary cements

Type I/I : Fine grain for cementation and luting –

Type I/II : Medium grain for bases, orthodontic purposes

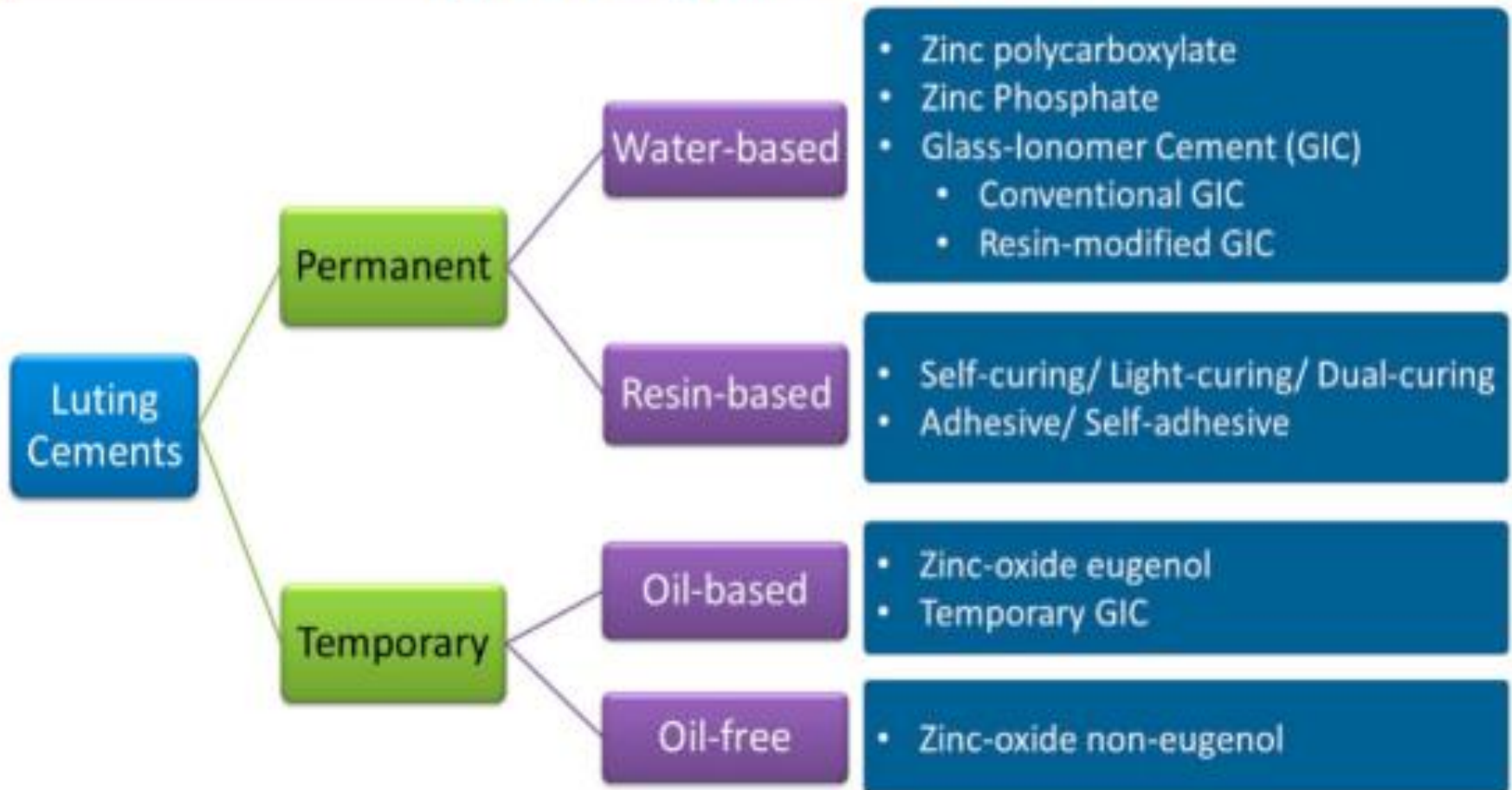
Type II: Restorative applications

Type III: Liner or base applications.

Cont`d

- Dental luting agents are cement that are applied to underlying tooth structures to secure a fixed prosthesis such as crowns, onlays, inlays and bridges.
- Luting agents are intended to fill the minute void between the tooth preparation (or implant abutment) and the restoration in place to prevent dislodgement during mastication.
- Luting agents may be classified as definitive (permanent) or provisional (temporary) according to their expected time required to stay inside the oral cavity.

Types of Luting Cements in Dentistry



Ideal requirements of cements

- Biocompatible with oral tissues
- Adequate mechanical characteristics
- Promote tissue health
- Adequate marginal seal
- Dissolution resistance
- Radi-opaque
- Aesthetically pleasant
- Cost-effective

Cont`d

Properties	Ideal Requirements
Biological	• Should be/have • Non-toxic and non-irritant. • Non-carcinogenic. • Should not cause any systemic reactions. • Should be cariostatic thus preventing secondary caries formation.
Chemical	• Should be chemically inert. • Solubility of the cement in oral fluids or any other fluids being taken by the patient should be negligible (maximum allowable solubility of cements in oral conditions is 0.2%). • Should bond chemically to the enamel and dentin. • PH should be neutral.
Rheological	Should be/have • Low film thickness to enable the easy flow of luting cement. • Longer mixing and working time. • Shorter setting time.

Cont`d

Mechanical	Should be/have • High compressive strength to withstand the masticatory forces. • High tensile strength to reduce the brittleness. • High modulus of elasticity. • Exhibit minimum dimensional changes on setting. • Restoration should take and retain a smooth surface finish. • Should bond chemically to the enamel and dentin.
Thermal	Should be/have • Good thermal insulator. • Coefficient of thermal expansion (COTE) should be similar to the tooth and artificial prosthesis.
Optical/aesthetic	• Should not alter the color of the tooth and artificial restorations/prosthesis. • Should have adequate radiopacity to enable detection of secondary caries and detection of incompletely filled cavities due to trapped air.
Miscellaneous	Should be /have • Easy to manipulate. • Inexpensive. • Longer shelf life.

Zinc phosphate cement

Zinc Phosphate Cement : Composition

Powder

Zinc Oxide	> 75 %	Main Constituents
Magnesium Oxide	13 %	Aids in sintering
Barium oxides	0.2 %	Radioopacity
Other oxides (Bismuth trioxide, Calcium oxide)	1.4 %	Smoothness of mix and fillers

Liquid

Phosphoric Acid	38 % – 59 %	Reacts with ZnO
Water	30% to 55%	Controls the rate of reaction
Alumuninium Phosphate	2% to 3%	Buffer
Zinc phosphate	(up to 10%)	

Zinc phosphate

Setting reaction	Mixing Time : 1.5 – 2 min Setting Time : 2.5 – 8 min Exothermic Reaction
Advantages	Highest elastic modulus; High Compressive Strength; Low film thickness and Low cost
Disadvantages	Acidic (possible pulpal irritation); Low tensile strength; High solubility (in oral fluids); brittle
Indications	Long span bridges Metal Crowns Metal-Ceramic Crowns Porcelain jacket crowns
Contra-indications	All-ceramic restorations – due to setting expansion

Zinc polycarboxylate cement

Zinc Polycarboxylate Cement : Composition

Powder

Zinc Oxide	89 %	Main Constituents
Magnesium Oxide	9 %	Aids in sintering
Barium oxides	0.2 %	Radioopacity
Other oxides (Bismuth trioxide, Calcium oxide)	1.4 %	Smoothness of mix and fillers

Liquid

Polyacrylic acid or a copolymer of acrylic acid	32 % – 48 %	Reacts with ZnO
Other carboxylic acids, such as itaconic acid or maleic acid	30% to 55%	Controls the rate of reaction
Stannous fluoride	-	adjust the setting time & increase the strength

Zinc poly carboxylate

Setting reaction	Mixing Time : 30 seconds – 2 min • Setting Time : 6 to 9 min • pH : rapidly rises from 3 to 6 • Pulpal response : mild Adhesion properties – mechanical & chemical (carboxyl group of tooth structure)
Advantages	Anti-bacterial; Low irritation; Adhesive to tooth structure; good compressive strength; Higher tensile strength than Zinc Phosphate
Disadvantages	Low pH initially; Low resistance to erosion in acidic environment and Short working time
Indications	Porcelain restorations Metal crowns Metal-ceramic crowns
Contra-indications	Titanium based restorations (cement discolouration occurs)

GLASS IONOMER CEMENT (GIC)

- Developed in the 1970s
- It is a tooth colored, anti-cariogenic restorative material
- Has combined properties of silicate cements and poly carboxylate cements

Classification:

Type I: Luting crowns, bridges, and orthodontic brackets

Type IIa: Esthetic restorative cements

Type IIb: Reinforced restorative cements

Type III: Lining cements, base



GIC

Glass Ionomer Cement : Composition

Powder

Silica (SiO) ₂	41.9	
Alumina (Al ₂ O ₃)	28.6	
Aluminium flouride (AlF ₃)	1.6	
Calcium flouride (CaF ₂)	15.7	
Sodium flouride (NaF)	9.3	
Aluminium phosphate (AlPO ₄)	3.8	
Barium/strontium oxide		radiopacity

Liquid

Polyacrylic acid or a copolymer of acrylic acid	40-50 %	Reacts with SiO and Al ₂ O ₃
Other carboxylic acids, such as itaconic acid or maleic acid	30% to 55%	Controls the rate of reaction
Tartaric acid		rate-controlling additive

GIC

Setting reaction	Setting Time : – Type I — 4-5 minutes and Type II — 7 minutes • Release of fluoride • Greater pulpal reaction than ZOE cement but less than zinc phosphate cement
Advantages	Adhere to teeth and metal; Fluoride release; Ease of Mixing Good flow; Cheap; Aesthetically pleasant; thermal compatible with enamel; Low shrinkage; Good resistance to acid dissolution
Disadvantages	Soluble in water; Rapid setting time; limitation especially in cementation of several units; Moisture sensitivity during setting Brittle; Low tensile strength; Poor wear resistance; Good resistance to acid dissolution
Indications	Metal and Metal-Ceramic Restorations; Porcelain restorations All Ceramic Crowns with high strength cores such as alumina or zirconium
Contra-indications	-----

Modifications to GIC

- Metal-reinforced GIC
- High-viscosity GIC
- Resin-modified GIC (hybrid ionomer)
- Calcium aluminate GIC
- Compomer

Resin modified GIC

- Resin-modified glass ionomers (RMGIC) were developed to improve the mechanical properties of GIC.
- The only difference in composition is the addition of a hydrophilic monomer called hydroxyethyl methacrylate resin (HEMA).
- These materials had a better fracture toughness and wear resistance compared to GIC as well as a higher moisture resistance and longer working-time

RMGIC

Setting reaction	Dual curing (chemical (self) and light curing
Advantages	Dual cure Fluoride release Higher flexural strength than GI Capable of bonding to composite materials
Disadvantages	Setting expansion may lead to cracking of all-ceramic crowns Moisture sensitive
Indications	Cavity liners; Core buildups; Luting cements Crowns
Contra-indications	All-ceramic crowns – due to uptake of water causing swelling and pressure on the crown; on Veneer – not retentive enough

Resin cement

- Resin cements are composed of resin matrix of bisgma or methacrylate and filler of five inorganic particles (20-80%) to ensure thin film thickness
- Resins bond micromechanically to prepared enamel, dentin, alloys and ceramic surfaces .
- Polymerization occurs and hardens these resin cement in place.
- Resin can be activated using a light cure, a chemical cure, or a dual-cure method. As the name implies, light-cure resin cement contains photo-initiator which can be activated by light energy.

Resin cement

Setting reaction	Dual curing (chemical (self) and light curing
Advantages	Strongest of the cement – highest tensile strength. Least soluble (in oral fluids) High micromechanical bonding to prepared enamel, dentin, alloys, and ceramic surfaces Neutral pH
Disadvantages	Setting shrinkage – contributing to marginal leakage Difficult sealing Requires a meticulous and critical technique
Indications	All crown types, Bonding fixed partial dentures; Inlays; Veneers Indirect resin restorations; Resin-fiber posts
Contra-indications	Metal crowns since it would not cure enough through the metal.

Temporary cements

- Zinc oxide euganol
- Zinc oxide non euganol

ZnE euganol

Composition	Supplied as two pastes or as a powder (zinc oxide) and liquid (zinc acetate, eugenol, olive oil)
Advantages	bactericidal effect due to free eugenol Excellent ability to seal the margins Radiopacity
Disadvantages	pulpal damage due to production of exotoxins high solubility in water interfere with polymerisation process and leads to discoloration so cant be used in patients considering permanent cementation using resin-based luting
Indications	Mainly used for lining under amalgam restorations and as a temporary crown restoration where prospective permanent restoration is nor resin based
Contra-indications	Thin dentin to protect the pulp Prospective resin based restoration

Zinc oxide non euganol

- The production of zinc oxide non-eugenol is formed by substituting the eugenol with an organic acid.
- Non-eugenol zinc oxide is recommended for patients who have post cementation hypersensitivity to eugenol and in patients considering permanent cementation using resin-based luting agents as the absence of eugenol eliminates the negative effects of it inhibiting resin polymerization.

Summary

Choice of luting agents in different clinical situations	
Clinical condition	Luting agent type preferred
Hypersensitive vital teeth	Zn PCB, ZoE
Hypersensitive teeth with average retention form	ZnPCB (only crowns and short-span bridges)
Hypersensitive teeth with more than average/excellent retention and resistance form and minimal dentin layer	ZoE
Long-span fixed partial dentures in areas of high masticatory stresses when abutment teeth are not sensitive	ZnP, GI, RMGI
Non-vital teeth with averages/less than average retention form (might have fractured cusp/wall)	GI, RMGI, Resin
Patients where complete isolation is difficult/in areas difficult to isolate (posterior mandibular region)	Resin
Patients with reduced saliva (xerostomia)	ZnPCB, RMGI
High caries index	GI, RMGI
Children with large pulp chambers	ZnPCB
Teeth prepared to receive partial veneer crown or retainer	Resin, RMGI

Steps of crown cementation

- Clean the preparation and crown with water spray.
- Air dry the preparation and isolate with cotton rolls.
- Checking fitting without cement first
- Mix cement according to manufacturer's instructions.
- Coat the fitting surface of the crown with cement.
- Seat the crown quickly with pressure.
- Wait for the setting time
- Remove the excess cement.
- Check the occlusion
- If occlusion is perfect Polish and finish

Occlusion verification

- Check for centric occlusion at supine and 90 degree sitting positions
- Check for maximum intercuspation
- Use articulation paper to check the occlusal discrepancies
- Remove the excess using high speed hand piece installed with round burs.

Cont`d



Complication of fixed prosthesis

- Get help from your neighbors and list as much as you can.....

Retainer/crown removal



The end

Thank you